

WHO consolidated guidelines for the management of common childhood illness

Management of asthma in children
and adolescents and bronchiolitis
in infants and young children



World Health
Organization

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Abbreviations

CI	confidence interval
CPAP	continuous positive airway pressure
CXR	chest X-ray
ECG	electrocardiogram
EMLc	WHO Model List of Essential Medicines for Children
FEV₁	forced expiratory volume in 1 second
GCS	Glasgow Coma Scale
GDG	Guideline Development Group
GINA	Global Initiative for Asthma
GRADE	Grading of Recommendations Assessment, Development and Evaluation
HFNC	high-flow nasal cannula
HHFNC	humidified high-flow nasal cannula
ICS	inhaled corticosteroids
ICS/LABA	inhaled corticosteroids/long-acting beta ₂ agonist
ICU	intensive care unit
IM	intramuscular
IMCI	Integrated Management of Childhood Illness
IV	intravenous
LABA	long-acting beta ₂ agonist
LMIC	low- and middle-income country
MART	maintenance and reliever therapy
MD	mean difference
MDI	metered-dose inhaler
NCD	noncommunicable disease
OR	odds ratio
PEF	peak expiratory flow
PICO	population, intervention, comparator and outcome
PICU	paediatric intensive care unit
RCT	randomized controlled trial
RR	relative risk
RRI	relative risk increase
RRR	relative risk reduction
RSV	respiratory syncytial virus
SABA	short-acting beta ₂ agonist
SD	standard deviation
SDG	Sustainable Development Goal
SFO	standard flow oxygen
SpO₂	peripheral capillary oxygen saturation
WHO	World Health Organization
WMD	weighted mean difference

Glossary

Apnoea: Temporary cessation of breathing, especially during sleep.

Asthma: A heterogeneous disease, usually characterized by chronic airway inflammation and episodes of airway narrowing and obstruction causing breathing difficulties.

Bilevel positive airway pressure: A type of non-invasive ventilation that provides two levels of pressure: higher during inhalation and lower during exhalation.

Bronchiolitis: An acute viral infection of the lower respiratory tract that typically affects infants and children aged less than 2 years.

Continuous positive airway pressure: A treatment method that uses mild air pressure to keep the airways open.

Cyanosis: A bluish discolouration of the skin due to low oxygen levels in the blood.

Humidified high-flow nasal cannula: A method for delivery of up to 100% humidified and heated oxygen at high flow rates of up to 60 litres per minute.

Hypertonic saline: A highly concentrated saline water solution used in nebulization to help relieve wheezing and breathing difficulty.

Hypoxaemia: Low levels of oxygen in the blood.

Moderate paediatric asthma exacerbation: A sudden increase in asthma symptoms, associated with: peak expiratory flow >50% best or predicted, increased respiratory rate (<30/min), increased heart rate (100–120/min), able to complete sentences in one breath and oxygen saturation (on air) 90–95%. These parameters are applicable to children aged ≥6 years.¹

Nebulization: A method of administering medication in the form of a mist inhaled into the lungs.

Non-invasive positive pressure ventilation: A method of oxygen delivery of positive pressure ventilation via a tight-fitting mask that covers the nose or both the nose and mouth without the need for endotracheal intubation.

Non-invasive ventilation: A method of providing ventilatory support without the need for endotracheal intubation.

Peripheral capillary oxygen saturation: A measure of the oxygen saturation level in the blood.

Prodrome: An early symptom indicating the onset of a disease.

Respiratory distress: Difficulty in breathing, often characterized by rapid breathing and use of accessory muscles.

Respiratory syncytial virus: A common virus that causes bronchiolitis and other respiratory infections in young children.

¹ For further details, see Table 2.

Severe paediatric asthma exacerbation: A sudden increase in asthma symptoms, associated with: peak expiratory flow $\leq 50\%$ best or predicted, increased respiratory rate ($>30/\text{min}$), increased heart rate ($>120/\text{min}$), use of accessory breathing muscles, inability to complete sentences in one breath, decreased oxygen saturation (on air) $<90\%$. These parameters are applicable to children aged ≥ 6 years.¹

Spacer: A device used with an inhaler to help deliver medication to the lungs more effectively.

Standard first-line therapy: The initial treatment of acute asthma with inhaled bronchodilators and systemic corticosteroids.

Tachycardia: Abnormally rapid heart rate.

Tachypnoea: Abnormally rapid breathing.

Ventilator-induced lung injury: Lung damage that occurs as a result of mechanical ventilation.

Wheezing: A high-pitched whistling sound made while breathing, often associated with respiratory conditions.



Summary of recommendations

Recommendations for the management of asthma in children and adolescents



Additional treatment to standard first-line therapy for paediatric asthma exacerbations

Recommendation 1a: Nebulized ipratropium bromide should be added to standard first-line therapy with inhaled short-acting beta₂ agonists (SABA) and systemic corticosteroids for severe paediatric asthma exacerbations in health facilities.

Recommendation 1b: Nebulized ipratropium bromide should be added for moderate paediatric asthma exacerbations not responsive to initial standard first-line therapy with inhaled SABA and systemic corticosteroids in health facilities.

Strong recommendation, low-certainty evidence.

Remarks

- Ipratropium bromide should be given in addition to standard inhaled or nebulized SABA and systemic corticosteroids in cases where children are not responding or when the severity of the exacerbation is extreme and impending collapse is likely.
- It is essential that first-level health care providers are trained and competent to assess the severity of an acute asthma exacerbation, start first-line therapy appropriately and monitor response.
- The results of the systematic review favoured the use of nebulized ipratropium bromide but could not confirm similar effects with administration via metered-dose inhaler (MDI), mask and spacer.



Second-line therapy options for paediatric asthma exacerbations

Recommendation 2: Intravenous (IV) magnesium sulfate (as a single short infusion) should be used as second-line therapy for severe paediatric asthma exacerbations, and for moderate exacerbations not responsive to standard first-line therapy with inhaled/nebulized medicines (SABA and ipratropium) and systemic corticosteroids in health care facilities.

Strong recommendation, low-certainty evidence.

Remarks

- Current evidence does not support the use of inhaled magnesium sulfate, and intramuscular (IM) magnesium sulfate has not been studied in acute asthma exacerbations.
- IV infusion of magnesium sulfate relaxes smooth muscle and may result in hypotension, a side-effect commonly seen with the use of IM magnesium sulfate in pregnant women.
- It is advised that magnesium sulfate should be diluted in a compatible solution and administered as a single IV infusion over 20 minutes.
- The most frequent dosing regime was 50 mg/kg of magnesium sulfate diluted in 100 mL normal saline with a maximum dose of 2 g administered over 20–30 minutes, although it has been suggested that higher doses up to 100 mg/kg and prolonged infusions may be beneficial life-threatening asthma.
- Regular blood pressure monitoring should be included in the routine observations of children receiving IV magnesium sulfate.

Recommendation 3: IV aminophylline should not be used as second-line therapy for severe paediatric asthma exacerbations in health facilities if alternative second-line options are available.

Conditional recommendation, low-certainty evidence.

Remarks

- The restricted use of aminophylline should be accompanied by the availability of appropriate inhaled SABA and systemic corticosteroids for the management of acute asthma exacerbation across all settings.
- In resource-restricted and rural settings aminophylline may be the only available agent for the management of acute asthma exacerbations.
- If the decision is made to use aminophylline, due to lack of alternatives, adequate monitoring and anti-emetics should be used to mitigate the risk of side-effects.

No recommendation: WHO makes no recommendation on the use of subcutaneous adrenaline in the management of acute asthma.

Remarks

- Adrenaline should be reserved for patients who present with symptoms and signs consistent with severe allergic reactions.
- The use of IM adrenaline for anaphylaxis is described in WHO's *Pocket book of hospital care for children* (1) and the *Integrated management of adolescent and adult illness* (2).

Recommendation 4: IV SABA should be used as a potential second-line therapy for severe paediatric asthma exacerbations in addition to standard first-line therapy with inhaled/nebulized medicines (SABA and ipratropium) and systemic corticosteroids in health care facilities.

Strong recommendation, high-certainty evidence.

Remarks

- IV SABA can be used as a potential second-line therapy, depending on the availability of other options. IV magnesium sulfate would be preferred as second-line therapy, if available (see Recommendation 5).
- IV SABA is most likely to be used in the PICU or settings where the infrastructure and monitoring already exist.
- Careful clinical monitoring is recommended to ensure the early identification and management of possible cardiovascular adverse effects, such as tachycardia, hypotension and arrhythmias.
- Where resources are available, it is advisable to have continuous electrocardiogram monitoring, and check serum electrolytes (hypokalaemia) and lactate.



Preferred second- and third-line therapy for paediatric asthma exacerbations

Recommendation 5: IV magnesium sulfate should be the preferred option for second-line therapy and IV SABA the preferred option for third-line therapy in severe paediatric asthma exacerbations not responding to standard first-line therapy.

Conditional recommendation, very low-certainty evidence.

Remarks

- A pragmatic stepwise approach to second-line therapies would be appropriate, and allow practitioners to choose the best possible option for their setting.
- In settings where neither preferred option is available, IV aminophylline may be considered as additional therapy in severe asthma exacerbations not responding to standard first-line therapy.



Combination maintenance and reliever therapy (MART) for the long-term management of asthma

Recommendation 6: Combination inhaler (steroid plus fast-onset long-acting beta2 agonist (LABA)) MART should be considered for children with persistent asthma symptoms requiring regular inhaled corticosteroid (ICS) treatment.

Conditional recommendation, moderate-certainty evidence.

Remarks

- Although evidence was considered for children aged 4–11 years, current formulations are licensed for children aged ≥ 6 years; hence, the recommendation is restricted to children aged 6–11 years.

- Current evidence supports the use of combination inhaler (steroid plus fast-onset LABA) MART for children with persistent asthma symptoms who are already taking or planning to start regular ICS therapy.
- The use of a combination inhaler results in a potentially simpler regime, which may be more easily followed by both patients and prescribers.
- Children with infrequent symptoms should use SABA as needed, considering the concomitant intermittent use of ICS.

Recommendations for the management of bronchiolitis in infants and young children



Guiding principle for using clinical symptoms and signs in supporting a probable diagnosis of bronchiolitis in an infant or child

Good practice statement 1: In infants and children aged less than 2 years presenting with cough and difficulty in breathing, a comprehensive history and assessment of clinical signs and symptoms should be done. A diagnosis of bronchiolitis can be suspected if they have a history of a prodrome of an upper respiratory tract infection with rhinitis or coryza and cough, followed by respiratory distress with a low-grade fever, and any presence of tachypnoea (fast breathing), chest indrawing and diffuse wheeze or crackles on auscultation.

Remarks

- Assessment of clinical signs should include the Integrated Management of Childhood Illness (IMCI) (3) general danger signs,² cyanosis, respiratory rate, looking for chest indrawing, listening for a wheeze or crackles, and measuring oxygen saturation.
- Infants with viral bronchiolitis are usually not very ill looking, and many have a prolonged expiratory phase of breathing associated with wheeze.
- During a respiratory syncytial virus (RSV) outbreak, a higher index of suspicion should be maintained in the differential diagnosis for infants presenting with respiratory symptoms.
- A comprehensive history should also elicit presence of any risk factors and comorbidities, such as prematurity, children who are immunocompromized, and children with asthma or reactive airway disease, chronic lung disease, chronic neurological conditions or congenital heart conditions, and breastfeeding status.

² The IMCI general danger signs are: inability to drink or breastfeed, vomiting everything, experiencing convulsions and being lethargic or unconscious.



Guiding principle for assessing infants and children with suspected bronchiolitis for hospital admission

Good practice statement 2: In infants and children aged less than 2 years presenting with suspected bronchiolitis, clinicians should perform a comprehensive assessment for presence of risk factors and signs of severe disease.

Indications for admissions include: Signs of severe disease, which include IMCI general danger signs,³ high temperature of $\geq 38^{\circ}\text{C}$, central cyanosis or peripheral capillary oxygen saturation (SpO_2) $< 90\%$, severe respiratory distress, apnoea or history of apnoea, and presence of risk factors for severe disease which include comorbidities,^a prematurity and age < 3 months.

Remarks

- The indications for admission found in the review are mostly similar to the current WHO guideline recommendations for the admission of severe pneumonia (1).
- Clinicians should avoid unwarranted admissions that may lead to exposure to nosocomial infections and avoidable hospital care costs.
- Local context should be taken into consideration, particularly the availability of follow-up or its accessibility, including travelling distances for patients, for example.
- The temperature threshold of $\geq 38^{\circ}\text{C}$ in cases of suspected bronchiolitis is significant and might be justification for admission or close observation.
- Comorbidities to consider in the history include: congenital cardiac lesions, chronic lung disease, cerebral palsy and neurological conditions.



Use of chest X-ray in the clinical management of bronchiolitis

Recommendation 1: In infants and children aged less than 2 years presenting or hospitalized with suspected bronchiolitis, routine use of chest X-ray is not suggested for the initial diagnosis and management of bronchiolitis.

Conditional recommendation, very low certainty of evidence.

³ The IMCI general danger signs are: inability to drink or breastfeed, vomiting everything, experiencing convulsions and being lethargic or unconscious.

Remarks

- Chest X-ray for hospitalized infants and children with suspected bronchiolitis should only be considered on a case-by-case basis; for example, in the following clinical situations:
 - if there is high suspicion for other alternate diagnosis or underlying cardiac or chronic lung disease related to risk factors elicited on history or physical examination;
 - if a child fails to respond to treatment or persistent symptoms for more than 2 weeks without improvement;
 - if there is suspected bacterial superinfection or recurrent episodes of pneumonia;
 - if there is severe respiratory distress or severe hypoxaemia requiring admission to high care or intensive care;
 - where a complication, such as effusion or pneumothorax, may be suspected; for example, when there are unilateral signs of severe dullness to percussion or absent breath sounds.



Administration of supplemental oxygen through nasal bubble continuous positive airway pressure (CPAP)

No recommendation: WHO makes no recommendation on the use of nasal bubble CPAP to administer oxygen rather than standard flow oxygen (SFO) through nasal prongs/face mask/headbox for better clinical outcomes.



Humidified high-flow oxygen versus SFO

Recommendation 2: In infants and children aged less than 2 years, hospitalized with acute bronchiolitis with hypoxaemia, humidified high-flow nasal cannula (HHFNC) oxygen is suggested rather than SFO. *Condition recommendation, low certainty of evidence.*

Remarks

- In most cases, children with acute bronchiolitis should be started on standard low-flow oxygen, which is very often effective in resolving the hypoxaemia.
- HHFNC oxygen is indicated where, despite standard low-flow oxygen therapy, the infant has persisting hypoxaemia (SpO₂ <90%) or severe respiratory distress.



Nebulized hypertonic (3%) saline versus no hypertonic saline

Recommendation 3: In infants and children aged less than 2 years presenting or hospitalized with bronchiolitis, a trial of nebulized hypertonic saline (3%) is suggested rather than no nebulization.

Conditional recommendation, low certainty of evidence.

Remarks

- The majority of studies used a comparison with normal saline which may underestimate the effect of hypertonic saline.
- A trial of nebulized hypertonic saline in an individual patient may aid in clearance of secretions, but might not always be effective.
- Health facilities experience overcrowding during peak RSV season, making any reduction in the length of hospital stays through the use of nebulized hypertonic saline especially valuable.
- Health care workers should be aware that hypertonic saline can induce bronchoconstriction and may often require co-administration with bronchodilators.

1. Introduction



Respiratory illness is a leading cause of morbidity among children globally, with both acute and chronic conditions contributing substantially to outpatient visits and hospital admissions throughout infancy, childhood and adolescence (4). Among the diverse clinical presentations of paediatric respiratory disease, wheeze is one of the most frequent and clinically important. Wheeze, defined as a high-pitched, musical sound produced by airflow through narrowed lower airways during expiration, is a common manifestation of multiple paediatric respiratory disorders.

In children aged less than 2 years, wheeze most commonly occurs in the context of acute viral bronchiolitis or other causes of small-airway obstruction. In older children and adolescents, wheeze is more often associated with asthma and other chronic airway diseases. Although these conditions differ in their underlying pathophysiology and natural history, they often present with similar clinical features, including wheeze, tachypnoea, chest indrawing and varying degrees of hypoxaemia, particularly during acute episodes.

Bronchiolitis typically affects infants in the first 2 years of life, with the highest incidence occurring in the first year (5). It is characterized by acute inflammation of the small airways, most often triggered by viral infection, leading to airway obstruction, increased work of breathing and impaired gas exchange. Asthma, in contrast, is a chronic inflammatory airway disease that manifests through recurrent episodes of wheeze, cough, chest tightness and breathlessness, with symptoms that vary in severity and over time.

Because bronchiolitis and asthma share overlapping clinical features especially wheeze and respiratory distress, distinguishing between them in routine clinical practice can be challenging, particularly in younger children and in settings with limited diagnostic capacity. These overlaps are most evident during acute presentations, when rapid decisions about treatment and referral must be made.

This guideline addresses two of the most common causes of wheeze and acute respiratory illness in childhood: asthma and bronchiolitis; and provides evidence-based recommendations to support accurate assessment, appropriate treatment and effective management across the age spectrum from infancy to adolescence.

1.1 Purpose and rationale

1.1.1 Asthma

Asthma control is frequently suboptimal, particularly in low-resource settings (6). Uncontrolled asthma symptoms lead to school absence and negatively affect children's educational and social opportunities (6), while emergency health care use and hospitalization impose substantial costs on families, health systems and the wider economy. In addition to the burden measured in disability-adjusted life years, asthma contributes significantly to avoidable morbidity and premature mortality (8). To make progress towards Sustainable Development Goal (SDG) target 3.4, which calls for a one third reduction in premature mortality from noncommunicable diseases (NCDs), it is essential to strengthen both acute and long-term asthma care, particularly in low- and middle-income countries (LMICs) where most asthma-related deaths occur (9). The importance of effective acute and chronic asthma treatment as priority interventions to reduce premature mortality has been underscored by initiatives such as NCD Countdown 2030 (10).

Current WHO guidance on asthma management is provided in the Pocket book of hospital care for children (1), which focuses primarily on acute care, and in the WHO Package of Essential Noncommunicable Disease (PEN) interventions for primary health care (11). However, there is a need to ensure that these recommendations reflect the current evidence base and are aligned with the WHO Model List of Essential Medicines for Children (EMLc), particularly in relation to medicines and treatment strategies for children and adolescents (12).

1.1.2 Bronchiolitis

Bronchiolitis is a common paediatric illness affecting children aged less than 2 years and is one of the leading causes of hospital admission in infancy. It is responsible for substantial increases in health care utilization, including oxygen use, respiratory support and inpatient care, resulting in significant medical costs and pressure on health systems. In LMICs, bronchiolitis contributes to child mortality through hypoxaemia, respiratory failure and complications of severe illness. To achieve SDG target 3.2, which aims to end preventable deaths of neonates and children aged less than 5 years, it is essential to improve the quality and consistency of bronchiolitis care, particularly in settings with limited resources.

At present, there are no stand-alone WHO guidelines dedicated specifically to the management of bronchiolitis, although the condition is addressed within the *Pocket book of hospital care for children* (1). There is also rapidly evolving evidence regarding key aspects of bronchiolitis management, including the role of nebulized hypertonic saline, high-flow nasal cannula (HFNC) oxygen therapy and continuous positive airway pressure (CPAP) in the escalation of care and prevention of respiratory failure (13). This evolving evidence base has not yet been clearly or consistently translated into global guidance. There is therefore a need to ensure that recommendations for bronchiolitis reflect current best evidence and are aligned with the WHO Model List of Essential Medicines for Children (12) and the *Interagency list of medical devices for essential interventions for reproductive, maternal, newborn and child health* (14).

Asthma and bronchiolitis account for a large proportion of paediatric respiratory morbidity, health care utilization and preventable mortality. Both conditions are prevalent and place substantial demands on health systems, and require timely, evidence-based management to improve outcomes. A WHO guideline that provides up-to-date, aligned, and context-appropriate recommendations for the management of asthma and bronchiolitis is therefore essential to support countries in achieving their child health and NCD targets under the SDGs.

2. Process and methods



The guidelines for the management of asthma in children and adolescents, and bronchiolitis in infants and young children were developed in accordance with the principles and procedures outlined in the *WHO handbook for guideline development* (15). The guideline development process followed WHO's standard steps, which include: (i) defining the general scope of the recommendations; (ii) identifying contributors and establishing the Guideline Development Group (GDG); (iii) formulating key questions and outcomes using the population, intervention, comparison and outcome (PICO) format; (iv) conducting systematic evidence reviews; (v) assessing the certainty of evidence using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach; (vi) formulating recommendations and drafting of the guideline narrative; (vii) reviewing and approving the recommendations through the GDG; (viii) conducting external peer review; (ix) submitting the guideline for review and approval by WHO's Guidelines Review Committee; and (x) finalizing and disseminating the final recommendations.

2.1 Contributors

A diverse group of contributors participated in the guideline development process, including clinical experts, methodologists, health system specialists and representatives from WHO headquarters and its regional and country offices. All contributors were selected based on their expertise and experience, and their roles were clearly defined in line with WHO policies to ensure effectiveness, transparency and minimize bias (Annex 1).

The WHO Steering Group comprised members from relevant technical units at WHO headquarters and WHO's regional and country offices. The WHO Steering Group provided technical oversight, coordinated the process and supported project management. This Group also established the general scope of the guideline, drafted potential key questions and identified and selected the GDG members.

The GDG included members from all WHO regions and a range of professional backgrounds. Their expertise covered the clinical management of asthma and bronchiolitis, as well as health systems, programme implementation and patient care. The GDG was supported by a guideline methodologist, who ensured methodological rigour throughout the process.

Systematic review teams were commissioned to synthesize the evidence for each key question. An External Review Group, composed of technical and implementation experts, reviewed the final draft of the guideline and provided feedback on content and presentation.

2.2 Management of conflict of interest

In accordance with the *WHO handbook for guideline development* (15), all individuals involved in the guideline development process, including members of the GDG, external reviewers and contributors, were required to disclose any potential conflicts of interest. These disclosures were reviewed by WHO staff and assessed for relevance and severity. Where a conflict of interest was identified, appropriate measures were taken to manage it, including exclusion from specific discussions or decisions, or full recusal from participation. The objective was to ensure transparency, maintain the integrity of the guideline development process and safeguard against undue influence. All conflict-of-interest declarations and management decisions were documented and are available upon request. Only individuals with no significant conflicts of interest were formally invited to join the GDG or External Review Group. Further details are provided in Annex 1.

2.3 Evidence retrieval and synthesis

At the initial two virtual GDG meetings, the GDG established the scope of the guideline; the key questions, using the PICO format; and identified key outcomes. Systematic reviews were commissioned to address each of the prioritized PICO questions (see annexes 2 and 3). These reviews adhered to Cochrane methods and standards and were conducted by independent review teams (16). The GRADE approach was used to assess the certainty of evidence for each critical and important outcome (17). Detailed methods are described in the respective systematic reviews (Web Annex (18)).

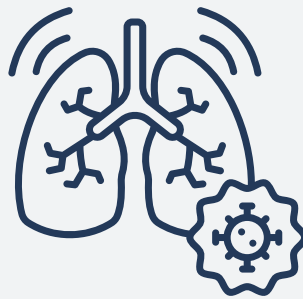
2.4 Formulation of recommendations

The GDG met virtually and in person between July 2023 and February 2024 to review the evidence and formulate recommendations. The meetings were chaired by a GDG member and facilitated by the guideline methodologist. The GDG used an explicit evidence-to-decision framework to guide discussions and ensure transparency in decision-making.

Recommendations were based on a balance of benefits and harms, the certainty of evidence, values and preferences, acceptability, feasibility, resource use and equity considerations. Each recommendation was classified as either strong or conditional, with:

- a **strong recommendation** indicating high confidence that the desirable effects outweigh the undesirable effects; and
- a **conditional recommendation** reflecting less certainty, acknowledging that the balance of desirable effects may vary on context or stakeholder values. The choices of individuals and policy or decision-makers will vary according to the relative values placed on the outcomes, acceptability, resources and other considerations.
- Consensus was the primary method for decision-making. All GDG members present had to agree on the final wording and the strength of each recommendation. If consensus could not be reached, a vote was planned as the fallback approach. Only GDG members participated in the decision-making process, although WHO staff, the methodologist and observers contributed to discussions.
- The GDG also identified research gaps during the formulation process. These reflect the uncertainties in the evidence base and highlight areas where further research could improve future guideline development and clinical decision-making. These research gaps are not prioritized or exhaustive but are intended to inform future research agendas.
- Following the GDG meeting, the guideline writer drafted the narrative sections, which were reviewed and approved by the GDG, and then submitted for external peer review by the External Review Group. After incorporating the feedback, the final recommendations were reviewed and approved by WHO's Guidelines Review Committee and prepared for publication and dissemination.

3. Management of asthma in children and adolescents



Asthma is a chronic respiratory illness, and one of the main NCDs included in WHO's *Global Action Plan for the Prevention and Control of Noncommunicable Diseases 2013–2030* (19). Every year, an estimated 426 000 people die from asthma, with half of these deaths in people aged less than 70 years. Over 95% of these premature deaths occur in LMICs (4). Asthma is a major cause of morbidity in young people and ranks among the leading causes of disability-adjusted life years worldwide (20). According to the Global Asthma Network Phase I study, about one in 20 school-aged children have severe asthma symptoms, with prevalence varying between and within countries (21).

The Global Initiative for Asthma (GINA) defines asthma as “a heterogeneous disease, usually characterized by chronic airway inflammation. It is defined by the history of respiratory symptoms, such as wheeze, shortness of breath, chest tightness and cough, that vary over time and in intensity, together with variable airflow limitation” (22).

Symptoms of asthma are typically worse at night and in the early morning, and are due to airway smooth muscle constriction, also known as bronchoconstriction, and inflammation of the airway lining. These symptoms are usually mild and may be controlled with medicine and avoidance of known allergens, but they can result in potentially life-threatening exacerbations.

Children and adolescents with asthma may experience short-term worsening of their asthma symptoms, with a progressive increase in wheeze, shortness of breath, chest tightness and cough. These episodes are known as acute asthma, attacks, flare-ups or exacerbations. In this document, the term exacerbation will be used. Acute asthma is commonly triggered by viral respiratory tract infections, exercise, change in weather and exposure to allergens (such as tree and grass pollen) and irritants (such as tobacco smoke or air pollution) (22).

Treatment aims to reduce airway smooth muscle constriction using bronchodilators and to reduce airway inflammation using corticosteroids. Inhaled bronchodilators and inhaled corticosteroids (ICS) are recommended for the long-term asthma management in the *WHO package of essential noncommunicable (PEN) disease interventions for primary health care* (11). During an acute episode, people with asthma need to change their usual treatment; for example, increasing their use of inhaled short-acting bronchodilators and taking a short course of systemic corticosteroids. More severe cases may require emergency treatment in a health facility and admission to hospital. However, effective asthma management involves a holistic approach that addresses both pharmacological and non-pharmacological management, as well as education and self-management aspects. Working in partnership with children and families is key in promoting good outcomes. Educating people with asthma on how to take the treatment effectively, how to avoid triggers, what their modifiable risk factors are, and the actions to take during acute exacerbations via personalized asthma action plans is essential.

3.1 Acute management of asthma

Inhaled bronchodilators are the first step in treating an acute asthma exacerbation. Short-acting beta₂ agonists (SABA) should be started as initial therapy at the onset of acute symptoms (cough, wheeze, difficulty breathing). Children with asthma should have access to a SABA inhaler and a plan of how to use it when an exacerbation occurs. A mild exacerbation can usually be managed at home if symptoms are relieved promptly with inhaled SABA.

If symptoms do not subside rapidly, patients are advised to seek medical review to assess the severity of the exacerbation. Assessment and classification of severity as mild, moderate, severe and life-threatening asthma depend on clinical signs including heart rate, respiratory rate, work of breathing, ability to talk, mental state and measuring oxygen saturation and peak expiratory flow (PEF) (if possible). The global and national guidelines available differ slightly in their classification of asthma exacerbation severity, although the overall clinical pictures described are broadly consistent (Table 1) (22–25).

Table 1 Classifications of asthma exacerbation severity in the available guidelines

	BTS/SIGN 2019 (23)	GINA 2023 (22)	National Asthma Council Australia (43)	NHLBI 2007 (25)
MODERATE	Speech	Able to talk in whole sentences		Dyspnoea interferes with or limits usual activity
	Oxygen saturation	≥ 92%	90–95%	>94%
	PEF rate	≥50% best or predicted	>50% best or predicted	>40% best or predicted
	Heart rate	≤140 bpm age 1–5 y ≤125 bpm age >5 y	100–120 bpm age ≥6 y	
	Respiratory rate	≤40/min age 1–5 y ≤30/min age >5 y	Increased	
SEVERE	Speech	Unable complete sentences in one breath or too breathless to talk or feed		
	Oxygen saturation	<92%	<90%	90–94%
	PEF rate	33–50% best or predicted	≤50% best or predicted	<40% best or predicted
	Heart rate	>140 bpm age 1–5 y >125 bpm age >5 y	>120 bpm age ≥6 y	
	Respiratory rate	>40/min age 1–5 y >30/min age >5 y	>30/min age ≥6 y	
	Other		Obvious respiratory distress	Dyspnoea at rest
LIFE-THREATENING	Exhaustion	Drowsy	Exhaustion	
	Hypotension			
	Cyanosis		Cyanosis	
	Silent chest	Silent chest	Silent chest	
	Poor respiratory effort		Poor respiratory effort	
	Confusion	Confusion		
	PEF <33% best or predicted			PEF <25% best or predicted
			Reduced consciousness	
				Too dyspnoeic to speak

bpm: beats per minute; BTS/SIGN: British Thoracic Society, Scottish Intercollegiate Guidelines Network; GINA: Global Initiative for Asthma; NHLBI: National Heart, Lung, and Blood Institute (United States of America); PEF: peak expiratory flow; y: years.

For patients with moderate or severe symptoms, prompt treatment with oral corticosteroids, for example, prednisolone or dexamethasone is advised. Administration of corticosteroids significantly reduces the need for hospital admission when given within 1 hour of presentation for emergency care and reduces relapse of symptoms (26, 27). In patients who are not able to tolerate oral medication, the intravenous (IV) or intramuscular (IM) route can be considered.

The initial treatment of acute asthma with inhaled bronchodilators and systemic corticosteroids is referred to in this document as standard first-line therapy.

3.2 Long-term management of asthma

The long-term management of asthma with inhaled bronchodilators and ICS has been standard practice for decades. However, evidence from large multi-centred RCTs, including mostly adults and older children, suggest that there should be changes in the way that these medicines are used (22); hence, the need to review the available evidence for children and adolescents.

3.3 Scope and target audience

This guideline reviews the current evidence to inform the recommendations for children and adolescents aged <19 years diagnosed with asthma. It is not intended to encompass all aspects of the acute management and the long-term management of asthma, but builds on existing recommendations. The recommendations on acute management relate to the emergency treatment of an asthma exacerbation in a health care facility, while the recommendations for long-term management relate to the ongoing treatment of the chronic condition with inhaled medicines, aiming at controlling symptoms and avoiding acute exacerbations. Therefore, for this guideline, the recommendations encompass the following topic areas:

- additional treatment to standard first-line therapy for paediatric asthma exacerbations;
- potential second-line therapy options for paediatric asthma exacerbations;
- preferred second- and third-line therapy for paediatric asthma exacerbations; and
- combination maintenance and reliever therapy (MART) for the long-term management of asthma.

The recommendations cover the options for inhaled and IV treatment in addition to standard first-line therapy with inhaled bronchodilators and systemic corticosteroids. They also take into consideration the need for early intervention with effective treatment to avoid potentially fatal outcomes, particularly in resource-limited settings where it may not be possible to escalate the level of care provided. The prioritized PICO questions discussed in section 4 are outlined in Annex 1.

4. Recommendations for the management of asthma in children and adolescents



4.1 Additional treatment to standard first-line therapy for paediatric asthma exacerbations

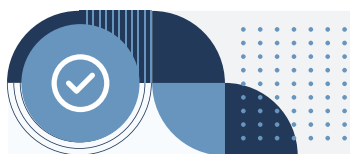
4.1.2 Inhaled or nebulized ipratropium bromide

Background

The inhaled anti-cholinergic, ipratropium bromide, is a quaternary ammonium derivative of atropine which reduces mucosal oedema and secretions and relieves cholinergic bronchomotor tone, resulting in bronchodilation. It is commonly used for the relief of bronchoconstriction due to limited absorption from the airways and minimal systemic effects (28). The antagonistic effect of ipratropium bromide on acetylcholine receptors in the airways has also been proposed to be particularly beneficial in viral infections, which cause acute asthma exacerbations by modifying these receptors (29). Inhaled ipratropium bromide is typically administered with an oxygen-driven nebulizer, but a metered-dose inhaler (MDI) with a mask and spacer is a commonly used alternative in low-resource settings where an oxygen supply is unavailable (30).

Ipratropium bromide is seldom used on its own and a review of six clinical trials in 171 children suggested that it is inferior to inhaled SABA for the management of acute asthma exacerbations in children aged older than 2 years (31). A combination of inhaled SABA and ipratropium bromide performed better than ipratropium bromide alone (treatment failure, odds ratio (OR): 2.65; 95% confidence interval (CI): –1.2 to 5.88) in this population (31). When nebulized ipratropium bromide was added to SABA in a review of 472 children aged 1–18 years hospitalized with acute asthma, no evidence of benefit for length of hospital stay (mean difference (MD): –0.28 hours; 95% CI: –5.07 to 4.52) or response to therapy was found (32). However, in a systematic review assessing combination inhalation of ipratropium bromide and SABA for the initial treatment of acute asthma in 2497 children, the addition of ipratropium bromide to SABA reduced the risk of hospital admission significantly (relative risk (RR): 0.73; 95% CI: 0.63–0.85) and a trend towards greater effect with increased severity was observed but did not reach statistical significance (33). No severe adverse effects were attributed to ipratropium bromide in any of the systematic reviews, confirming its favourable safety profile.

The evidence supporting ipratropium bromide as an effective and safe bronchodilator, and its use in combination with SABA inhalations, requires further investigation to define its use as an additional treatment in children who present with acute asthma exacerbations.



Recommendation 1a:

Nebulized ipratropium bromide should be added to standard first-line therapy with inhaled SABA and systemic corticosteroids for severe paediatric asthma exacerbations in health facilities.

Strong recommendation, low-certainty evidence.

Recommendation 1b:

Nebulized ipratropium bromide should be added for moderate paediatric asthma exacerbations not responsive to initial standard first-line therapy with inhaled SABA and systemic corticosteroids in health facilities.

Strong recommendation, low-certainty evidence.

Remarks

- Ipratropium bromide should be given in addition to standard inhaled or nebulized SABA and systemic corticosteroids in cases where children are not responding or when the severity of the exacerbation is extreme and impending collapse is likely.
- It is essential that first-level health care providers are trained and competent to assess the severity of an acute asthma exacerbation, start first-line therapy appropriately and monitor response.
- The results of the systematic review favoured the use of nebulized ipratropium bromide but could not confirm similar effects with administration via MDI, mask and spacer.

Justification

The review found, with moderate-certainty evidence, that less hospitalization (relative risk reduction (RRR): 5.2%) is needed when nebulized ipratropium bromide is used in addition to standard first-line therapy. However, when inhaled ipratropium bromide was compared to placebo, the hospitalization rate was similar (very low-certainty evidence), but a cost benefit is likely (34-36). The GDG considered that length of hospital stay may be influenced by various aspects of health care delivery (timing of ward rounds, availability of a pharmacist) and is not solely dependent on the underlying disease process. There was significant variability in the methods used to quantify improvements in symptoms, and low to very low evidence that it may favour ipratropium bromide. Additional literature considered suggests that most providers and caregivers value a reduction in hospitalization and probable improvement in asthma symptoms over a small risk of non-severe adverse effects and that the use of ipratropium bromide is cost-effective when used for severe exacerbations (37).

Due to the potentially life-threatening nature of acute asthma exacerbations, early intervention is imperative to treatment success. Therefore, given the moderate desirable effects, the small increase in non-severe adverse events, and values, acceptability, equity and feasibility which all favour the use of ipratropium bromide, the GDG made a strong recommendation in favour of the use of ipratropium bromide.

Summary of evidence

The evidence reviewed included 24 studies of 3238 children aged 18 years and younger with acute asthma exacerbation, as defined by the study criteria. Four studies were conducted in lower-middle-income countries (India, Pakistan) (38-41) and three in upper-middle-income countries (Mexico, Thailand, Türkiye) (42-44). Children were recruited from emergency and outpatient departments and all children received and failed to improve on standard first-line therapy, including at least three inhalations of SABA and systemic corticosteroids. The addition of ipratropium bromide to standard first-line therapy was compared to standard first-line therapy alone or in addition to placebo. Ipratropium bromide was typically administered as a nebulization within the first hour after presentation, but the dosing varied considerably. The critical outcomes under investigation included hospitalization, escalation of care (mechanical ventilation or paediatric intensive care unit (PICU) admission), length of hospital stay, morbidity (symptom improvement) and adverse effects.

Compared to placebo, the group receiving nebulized ipratropium bromide had a reduced need for hospitalization (RR: 0.76; 95% CI: 0.64–0.9, moderate-certainty evidence) (38, 42, 43, 45-49), but there was no difference in PICU admissions (RR: 0.91; 95% CI: 0.35–2.32, low-certainty evidence) (48, 50) or length of hospital stay in hours (MD: 1.75; 95% CI: –0.87 to 4.36, high-certainty evidence) (51-54) between groups. One study included children with moderate asthma exacerbation and compared inhaled ipratropium bromide administered with an MDI to placebo (RR: 1.09; 95% CI: 0.94–1.27) and did not show a benefit in hospitalization (very low-quality evidence) (55). Improvements in symptoms were measured using PEF measurement differences after interventions in seven trials (MD: 8.56; 95% CI: 1.24–15.88, very low-certainty evidence) (40-42, 47, 53, 56, 57); forced expiratory volume in 1 second (FEV₁) % predicted in three trials (MD: 15.64; 95% CI: –7.49 to 38.77, very low-certainty evidence) (40, 56, 58); and clinical asthma score improvement (MD: –0.38; 95% CI: –0.63 to –0.12, low-certainty evidence) in eight trials (39, 40, 42, 47, 51, 53, 56, 57). Adverse events included nausea, vomiting, headaches, dry mouth, abdominal pain and conjunctivitis and occurred in both groups (RR: 1.39; 95% CI: 0.95–2.05, moderate-certainty evidence) (38, 43, 45, 46, 49, 52, 55, 58).

Benefits and desirable effects

The addition of nebulized ipratropium bromide to standard acute asthma therapy reduced the hospitalization (RRR: 5.2%) of children presenting with asthma exacerbation. However, the addition of ipratropium bromide did not reduce the need for PICU admission (RR: 0.91) or length of hospital stay. The GDG considered that the length of hospital stay may be influenced by various aspects of health care delivery (timing of ward rounds, availability of a pharmacist) and is not solely dependent on the underlying disease process. An improvement in symptoms was evaluated by several proxy measures including a significant improvement seen in PEF (MD: 8.56) and reduction of clinical asthma severity scores (MD: –0.38), but there was no meaningful difference in FEV₁ % predicted (MD: 15.64) between the groups. These measurements were considered too heterogeneous to influence the recommendation. The GDG also noted that the greater effect of ipratropium bromide and SABA synergism with increased asthma exacerbation severity previously observed was not evaluated by the systematic review but may be relevant.

The GDG judged the desirable effects of ipratropium bromide in addition to standard first-line therapy in acute asthma exacerbations as moderate (low-certainty evidence).

Harms and undesirable effects

Minor adverse effects were observed equally in both the ipratropium bromide and placebo groups (RR: 1.39) and the GDG therefore judged the undesirable effects associated with ipratropium bromide to be small (moderate-certainty evidence).

Other considerations

Preferences, values and acceptability: The GDG agreed that most providers and caregivers would value a reduction in hospitalization and probable improvement in asthma symptoms over a small risk of minor adverse effects. This view is supported by a qualitative study of parents and caregivers of children presenting to hospital with a severe acute exacerbation of asthma in 12 countries (high- and middle-income) (37). Caregivers highlighted length of hospital stay, intensive care unit (ICU) or high dependence unit admission, and treatment costs as the most important outcomes. The GDG considered that a reduction in hospitalization would equate to a reduced hospital stay.

Resources required: Several studies have previously evaluated the cost and cost-effectiveness of ipratropium bromide in addition to standard acute asthma management in LMICs (34-36). These studies suggest that the provision of ipratropium bromide in addition to inhaled SABA delivered using a mask and spacer device is more beneficial financially than providing the same drugs in nebulized form (35, 36). Additionally, the cost benefit is attributed to a reduction in hospitalization and is observed when ipratropium bromide is reserved for severe exacerbations (34). The GDG was concerned that the provision of ipratropium bromide using an oxygen-driven nebulizer would incur additional costs to the health care system and/or caregivers, but considered that it could be cost-effective (moderate savings) in specific circumstances, probably favouring the intervention. The GDG also recognized that inappropriate use of ipratropium bromide in addition to poorly administered standard first-line therapy, may incur additional cost without much benefit.

Equity: The GDG considered that increased drug and equipment costs could contribute to inequity and low-resourced settings would be unfairly challenged by the required use of these interventions. The GDG decided that by advocating for access to ipratropium bromide through its inclusion on the EMLc, equitable access to the drug and its beneficial effects would be increased (12). The GDG considered that inclusion of a drug on the EMLc is needed to stimulate national procurement and increased demand is most likely to drive down local costs.

Feasibility: The GDG acknowledged that additional equipment would be necessary for implementation, and that masks and spacer devices are not always available, and agreed that it would only be feasible to implement this intervention at all levels if the equipment is procured. The GDG agreed that this intervention, if included on the EMLc, would be acceptable to all key stakeholders. In addition, the additional cost associated with using oxygen-driven nebulized ipratropium bromide would limit the implementation of this intervention and disadvantage resource-limited settings where oxygen is not available.

Implementation considerations

- Ipratropium bromide will need to be included on the EMLc, and equipment for nebulization should be included on the essential equipment list.
- To lower the cost of ipratropium bromide administration, use of an MDI, mask and spacer, which is as effective as nebulized therapy, should be considered in settings where oxygen and the equipment for nebulization are not immediately available.
- A 500 mL bottle spacer is an effective alternative to a conventional commercial spacer, but should be constructed following the methods that have been demonstrated as effective (59-61).

- In severe asthma, children should receive frequent doses of nebulized bronchodilators driven by oxygen, and, after improvement, treatment can be adjusted to MDI as tolerated (30).
- There is an opportunity to tailor the intervention based on the severity of the exacerbation and the availability of equipment.
- Once clinically stable and ready for discharge, patients and families must have a clear understanding of the treatment to be continued at home, how to monitor symptoms and signs, and action to take in case of a deterioration.
- Educating families on asthma self-management improves outcomes in children, including a reduction in school absence and emergency care visits (62).
- Patients should be provided with written action plans to guide home management of asthma.
- Asthma action plans based on monitoring of symptoms, rather than peak flow measurements, are preferred by children and are more likely to prevent acute care visits (63).

Research gaps

- An analysis of the failure rate of standard acute asthma therapy with inhaled SABA and corticosteroids.
- Investigating the effect of dose, timing, route of administration and duration of ipratropium bromide therapy on the effectiveness of the intervention.
- Investigating the ideal treatment options for children discharged after an acute asthma exacerbation who received ipratropium bromide.
- A post-implementation analysis of the effect of ipratropium bromide on referrals to tertiary centres in LMICs.

4.2 Second-line therapy options for paediatric asthma exacerbations

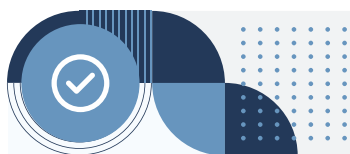
4.2.1 IV magnesium sulfate

Background

Asthma is a chronic lung disease characterized by airway inflammation and hyperresponsiveness of airway smooth muscles. There is growing evidence that magnesium sulfate (MgSO_4) may have a role in managing asthma through its dual effect as an anti-inflammatory and bronchodilating agent (64, 65). Bronchodilatation is attributed to the blockade of calcium channels in the airway smooth muscles and a reduction in airway excitability. This anti-inflammatory process is due to the eradication of oxygen free radicals, the stabilization of T-cells and mast cells, and the facilitation of endogenous nitric oxide and prostaglandin release (64, 66). Additionally, magnesium sulfate acts as a beta-receptor agonist, resulting in a synergistic effect when administered with salbutamol (67).

Previous research and evidence reviews have shown that administering magnesium sulfate via IV or inhaled routes can help manage acute exacerbations of asthma. Both routes of administration are fast-acting and effective, although the IV route has greater bioavailability but with associated with systemic side-effects (68). Intravenous magnesium sulfate is currently used as adjunctive therapy for severe asthma exacerbations. Its addition to standard asthma therapy in both children and adults has been shown to reduce hospitalizations and clinical asthma scores (65, 69). However, it is acknowledged that the evidence in the literature has provided conflicting results, with studies exploring both the IV and nebulized routes of administration (70-72). In the most recent review of three RCTs in 816 children aged less than 5 years with acute asthma exacerbation, nebulized or IV magnesium sulfate did not show any benefit in hospitalizations or clinical outcomes when compared to standard treatment (73).

Despite the conflicting evidence, several asthma treatment guidelines currently recommend giving IV magnesium sulfate to children having asthma exacerbations who have not responded well to other treatments, but it remains unclear whether it is safe and similarly effective for children. The evidence review of IV magnesium sulfate to date is inconclusive and warrants further investigation.



Recommendation 2:

IV magnesium sulfate (as a single short infusion) should be used as second-line therapy for severe paediatric asthma exacerbations, and for moderate exacerbations not responsive to standard first-line therapy with inhaled/nebulized medicines (SABA and ipratropium) and systemic corticosteroids in health care facilities.

Strong recommendation, low-certainty evidence.

Remarks

- Current evidence does not support the use of inhaled magnesium sulfate, and IM magnesium sulfate has not been studied in acute asthma exacerbations.
- Intravenous infusion of magnesium sulfate relaxes smooth muscle and may result in hypotension, a side-effect commonly seen with the use of IM magnesium sulfate in pregnant women.
- It is advised that magnesium sulfate should be diluted in a compatible solution and administered as a single IV infusion over 20 minutes.
- The most frequent dosing regime was 50 mg/kg of magnesium sulfate diluted in 100 mL normal saline with a maximum dose of 2 g administered over 20–30 minutes, although it has been suggested that higher doses up to 100 mg/kg and prolonged infusions may be beneficial life-threatening asthma.
- Regular blood pressure monitoring should be included in the routine observations of children receiving IV magnesium sulfate.

Justification

The GDG judged severe asthma as a life-threatening condition for which early and effective treatment is necessary to avoid adverse outcomes and therefore made a strong recommendation in favour of IV magnesium sulfate, while acknowledging contextual differences in how this will be implemented. Given the desirable benefits, trivial non-severe adverse effects and values, acceptability, equity and feasibility, which all favoured the use of IV magnesium sulfate, the GDG made a recommendation in favour of IV magnesium sulfate as second-line therapy for severe paediatric asthma exacerbations, and for moderate exacerbations not responsive to standard first-line therapy with inhaled SABA and systemic corticosteroids. The GDG was mindful that IV access and the infusion pumps required for IV magnesium sulfate may result in additional costs in settings where equipment such as infusion pumps and monitoring are unavailable. The GDG supports a tailored approach in these settings where IV magnesium sulfate is reserved for children with more severe exacerbations.

Summary of evidence

This meta-analysis included nine studies of 473 children aged 18 years and younger with acute asthma exacerbation as defined by the study criteria and included upper-middle-income countries (Argentina, Brazil, Paraguay, Türkiye) (74-77) and one lower-middle-income country (India) (78). Children were recruited from emergency departments, and studies used various criteria for defining the severity of the exacerbations. In four trials, children with moderate to severe exacerbations were included based on a PEF of <60–70% (74, 79, 80) or a pulmonary index score >8 (81); in four trials (76-78, 82) inclusion was based on a variety of clinical scoring systems (83-85); and in one trial inclusion was based on clinical symptoms as per local guidance (75). All children received and failed to improve with standard first-line therapy, including at least three inhalations of SABA and systemic corticosteroids. The addition of magnesium sulfate to standard first-line therapy was compared to standard first-line therapy alone or in addition to placebo. Magnesium sulfate was typically administered as a short IV infusion diluted in normal saline and given over 20 minutes.

The critical outcomes under investigation included hospitalization, escalation of care (mechanical ventilation or PICU admission), length of hospital stay, morbidity (symptom improvement) and adverse events. When compared to placebo, the need for hospitalization was less among the magnesium sulfate group (RR: 0.70; 95% CI: 0.54–0.90, low-certainty evidence) (79-81) and the need for non-invasive ventilation (RR: 0.17; 95% CI: 0.05–0.54) favoured magnesium sulfate (76). However, escalation of care to invasive mechanical ventilation was similar in both groups (RR: 0.35; 95% CI: 0.11–1.17, very low-certainty evidence) (75, 76, 82), and there was no difference in PICU admission rate (RR: 0.58; 95% CI: 0.15–2.22, very low-certainty evidence) (82) or the length of hospital stay in hours between the two groups (MD: –78.86; 95% CI: –200.37 to 42.66, low-certainty evidence) (75, 76, 78, 82). One study used PEF as a proxy for improvement in symptoms and reported an improvement with IV magnesium sulfate (MD: 36.60; 95% CI: 33.28–39.92) (74). When clinical asthma severity scores were used, no difference was found (MD: –0.18; 95% CI: –1.35 to 0.98, low-certainty evidence) (81, 82). Additional SABA nebulizations were similar in both groups (MD: 0.14; 95% CI: –0.53 to 0.80, low-certainty evidence) (80, 82). The included studies did not report enough data to enable meta-analysis; however, minor side-effects reported included epigastric warmth (12.5%), pain (16.6%), and tingling and numbness (12.5%) at the site of infusion (78, 82). No serious adverse events were reported in any of the included studies.

Benefits and desirable effects

The GDG determined that admission to PICU and escalation to ventilation should be considered priority outcomes. The GDG also recognized that most children with acute asthma exacerbation who receive magnesium sulfate have severe symptoms and are likely to require hospitalization. There was very low-certainty evidence that the addition of magnesium sulfate to standard first-line therapy reduced the need for non-invasive ventilation (RR: 0.17), ventilatory support overall (RRR: 5.8%) and decreased hospitalizations (RRR: 23%). Based on this evidence, the GDG judged the desirable effects as moderate.

Harms and undesirable effects

Minor adverse effects such as facial flushing, epigastric warmth, pain and tingling and numbness at the infusion site were reported in two studies. The GDG judged these undesirable effects to be trivial.

Other considerations

Preferences, values and acceptability: The GDG agreed that most caregivers would value the avoidance of mechanical ventilation and PICU admission over the trivial risk of non-severe adverse effects attributed to magnesium sulfate administration. This view is supported by a qualitative study of parents and caregivers of children presenting to hospital with a severe acute exacerbation of asthma in 12 countries (high- and middle-income) (37). Caregivers represented in this study highlighted length of hospital stays, ICU or high dependence unit admission and treatment costs as the most important outcomes. There was agreement among the GDG that the intervention would be acceptable to key stakeholders, with the provision that contextual differences would be considered.

Resources required: The systematic review did not include studies providing a cost-effectiveness analysis, but previous studies have shown a cost benefit based on higher quality-adjusted life years and the avoidance of hospitalization with a number needed to treat of four when IV magnesium sulfate was used in acute asthma exacerbation (86, 87). The GDG concluded that using IV magnesium sulfate in acute asthma exacerbation may result in moderate savings to the health care system. However, due to the additional cost incurred for the administration equipment and staff required to administer IV magnesium sulfate as a slow bolus (over 20 minutes) this may preclude its use from resource-limited settings.

Equity: The GDG also decided that the increased use of magnesium sulfate for children with asthma exacerbation is unlikely to affect its availability for pregnant women with (pre)eclampsia as the numbers treated would be too small to have an impact. The GDG concluded that health equity would be increased by the intervention.

Feasibility: The GDG recognized that the implementation of IV magnesium sulfate would be difficult in resource-limited settings where infusion pumps are not routinely available. In these settings, the GDG suggested that IV magnesium sulfate may be reserved for the treatment of more severe cases. In addition, IM magnesium sulfate for the management of (pre)eclampsia in pregnancy is included in the WHO Model List of Essential Medicines and is widely available (88, 89).

Implementation considerations

The GDG judged the implementation of the intervention as probably feasible.

Research gaps

- Investigating higher doses and prolonged magnesium sulfate infusions in the management of acute asthma exacerbations.
- Investigation of IM magnesium sulfate as an alternative route of administration in children with acute asthma exacerbation in resource-limited settings where IV access is not attainable.

4.2.2 IV aminophylline

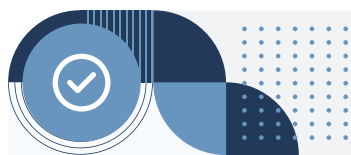
Background

Aminophylline is a methylxanthine bronchodilator composed of theophylline and ethylenediamine. Theophylline relaxes the smooth muscle of the bronchial airways and pulmonary blood vessels and reduces airway responsiveness to histamine, methacholine, adenosine and other chemical mediators. Aminophylline causes non-selective modulation of intracellular calcium release, inhibition of phosphodiesterase and antagonism of the adenosine receptor (90). Furthermore, aminophylline stimulates endogenous catecholamine release resulting in inotropic and chronotropic effects on myocardium (91). Aminophylline has a narrow therapeutic range (10–20 mg/dL) and severe toxicity may occur at higher levels, resulting in tachycardia, diuresis, hypokalaemia, convulsions and sudden death due to cardiac dysrhythmia (92).

The addition of IV aminophylline to standard first-line therapy with SABA and systemic corticosteroids remains controversial, and treatment guidelines vary from country to country (22, 93, 94). One meta-analysis concluded that the addition of IV aminophylline improved lung function within 6 hours of treatment but reduction in symptoms, number of nebulized SABA doses or length of hospital stay, but was associated with significant increased risk of nausea and vomiting (95, 96). More recently, in a review of IV aminophylline use in adults, IV aminophylline had no beneficial effects over standard of care and resulted in a 15% increase in arrhythmias or palpitations (97).

Internationally, there has been a shift away from using IV aminophylline in acute asthma exacerbations and experts have cautioned against its continued use (22). Despite this, aminophylline is widely available, affordable and remains part of the limited repertoire of drugs for the management of acute asthma exacerbations in remote areas (98).

Based on the current available evidence and practices, there is need to define the role of IV aminophylline in the management of acute asthma exacerbation.



Recommendation 3:

IV aminophylline should not be used as second-line therapy for severe paediatric asthma exacerbations in health care facilities if alternative second-line options are available.

Conditional recommendation, low-certainty evidence.

Remarks

- The restricted use of aminophylline should be accompanied by the availability of appropriate inhaled SABA and systemic corticosteroids for the management of acute asthma exacerbation across all settings.
- In resource-restricted and rural settings, aminophylline may be the only available agent for the management of acute asthma exacerbations.
- If the decision is made to use aminophylline, adequate monitoring and anti-emetics should be used to mitigate the risk of side-effects.

Justification

Given the trivial benefits, moderate adverse events and lower acceptability, the GDG suggested against the use of aminophylline in addition to first-line therapy in acute severe asthma exacerbations. This was driven by the low certainty of evidence that there no benefits associated with IV aminophylline, in addition to standard first-line therapy for acute asthma exacerbation, in the length of hospital stay, number of additional nebulized SABA doses needed during the study duration or improvement in asthma severity scores. There is moderate-to-high certainty of evidence that the IV aminophylline group had a moderate increase in undesirable adverse effects in the form of nausea and vomiting. The GDG recognized that there are settings in which aminophylline is the only available option and that there may still be a place for aminophylline as a second-line agent when appropriately monitored.

Summary of evidence

The evidence reviewed included nine studies of 464 children aged 18 years and younger with acute asthma exacerbation, as defined by the study criteria, mostly conducted in high-income settings, except for two upper-middle-income countries (Brazil, Türkiye) (99-101). Two trials (102, 103) scored severity based on the pulmonary index, four studies on standardized asthma severity scores (99, 104-106), two studies based on self-authored asthma severity score (100, 107), two studies used the Wood-Downes clinical scoring system (85) and the other study used a lack of clinical response to standard first-line asthma therapy (101). All children received and failed to improve with standard first-line therapy, including at least three inhalations of SABA and systemic corticosteroids. The addition of IV aminophylline to standard first-line therapy was compared to standard first-line therapy alone or in addition to placebo. The dosing and administration of aminophylline varied, but typically included a loading dose of diluted aminophylline administered as a short bolus over 20–30 minutes, followed by an infusion of diluted aminophylline. One study administered aminophylline boluses 6-hourly (101) and three studies calculated the aminophylline infusion rate based on therapeutic drug monitoring (102, 103, 106). The critical outcomes included the length of hospital stay, morbidity (symptom improvement) and adverse effects.

When compared to placebo, the IV aminophylline group showed no significant difference in the length of hospital stay in hours (MD: -3.62; 95% CI: -13.05 to 5.82) (101, 103, 105, 106), number of SABA nebulizers needed during the study duration (MD: 0.25; 95% CI: -0.39 to 0.89) (99, 101, 103, 105, 107), asthma severity score (MD: 0.13; 95% CI: -0.79 to 1.06) (99, 106), percentage improvement of predicted FEV₁ (MD: 7.00; 95% CI: -7.55 to 21.55) (103), improvement in PEF after 1 hour (MD: -0.14; 95% CI: -0.32 to 0.04) (105) or the need and duration of oxygen supplementation in hours (MD: -1.90; 95% CI: -4.37 to 0.57) (101). Adverse effects observed in the IV aminophylline group included emesis (RR: 3.52; 95% CI: 2.04–6.06) (99, 102, 104, 105, 107) and nausea (RR: 4.92; 95% CI: 2.41–10.06) (102, 104, 105). Other adverse effects such as headache (RR: 1.49; 95% CI: 0.52–4.25) (102, 104, 105), tremors (RR: 1.35; 95% CI: 0.88–2.06) (105, 107), abdominal pain (RR: 6.0; 95% CI: 0.31–115.6) (104), arrhythmia (RR: 0.40; 95% CI: 0.02–9.12) (104), irritability (RR: 1.16; 95% CI: 0.70–1.95) (104) and seizures (RR: 1.01; 95% CI: 0.06–15.91) (105) were observed, but no difference was detected between groups.

Benefits and desirable effects

There was low-certainty evidence that the use of IV aminophylline had no significant impact on the critical outcome of reduction in the length of hospitalization (MD: -3.62 hours), improved asthma severity scores (MD: 0.13) or the number of additional doses of nebulized SABA (MD: -1.9). These findings are consistent with previous reviews of IV aminophylline use in children with acute asthma exacerbation, and the GDG judged the desirable effects to be trivial (96, 108).

Harms and undesirable effects

An increase in both nausea (relative risk increase (RRI): 23.1%) and vomiting (RRI: 8.3%) in the groups receiving aminophylline were judged by the GDG as moderate undesirable effects that are best avoided in children with acute asthma exacerbation. The GDG considered that anti-emetics routinely administered in settings where aminophylline is still in use may mitigate these adverse effects. However, it was also noted that newer, appropriate and effective anti-emetics with fewer side-effects in children are not routinely available in resource-limited and remote settings where aminophylline is likely to be used (109). Major side-effects such as arrhythmias and seizures were only reported in one study which found no difference between groups.

Other considerations

Preferences, values and acceptability: The GDG determined that most providers and caregivers would want to avoid the adverse effects of aminophylline. This view is supported by a review of antiemetic use in children, indicating that nausea results in more frequent hospitalization, and would preferably be avoided by families (109). The GDG agreed that a recommendation against the use of aminophylline would be acceptable in most settings but recognized that practitioner views on the use of IV aminophylline are often entrenched with personal choice and locally accepted practices determining its use (110). The GDG also recognized that the continued use of aminophylline due to the unavailability of alternatives will continue and should be addressed.

Resources required: No studies evaluated cost or cost-effectiveness, but aminophylline is known to be widely available and cheap.

Equity: In view of the lack of evidence supporting the use of aminophylline in acute asthma exacerbation, the GDG agreed that cautioning against its use and advocating for the availability of preferable drugs would improve access to better asthma treatment and improve equity. The ongoing use of aminophylline in resource-limited settings where access to alternatives is not available decreases equitable management of asthma exacerbations in those populations.

Feasibility: The GDG agreed that it is feasible to limit the use of aminophylline in acute asthma exacerbations but recognized that there may be a role for aminophylline as a second-line agent in settings where preferred alternatives are not available. This reality should not distract from advocating for the use of more efficient drugs in asthma exacerbation management.

Implementation considerations

- The restricted use of aminophylline should be accompanied by ensuring availability of inhaled SABA and systemic corticosteroids for the management of acute asthma exacerbation across all settings.
- It is important that a recommendation against the use of IV aminophylline should not disadvantage resource-restricted and rural settings where aminophylline may be the only available agent for the management of acute asthma exacerbations.
- If the decision is made to use aminophylline, adequate monitoring and anti-emetics should be used to mitigate the risk of side-effects.
- Children receiving aminophylline require clinical monitoring including heart rate, respiratory rate, electrocardiogram (ECG) and neurological observations.
- Where available, blood gases and serum aminophylline/theophylline concentrations should be considered in those already receiving oral therapy and on prolonged treatment (23).

Research gaps

Important research priorities include the development of an internationally agreed core outcome set for future trials in acute severe asthma exacerbations and the determination of clinically important differences in these outcomes, which can then inform adequately powered future trials on the benefit of:

- the use of aminophylline in avoidance of PICU admission for severe paediatric asthma exacerbation; and
- the use of aminophylline as rescue therapy in children admitted to PICU with severe asthma exacerbation.

4.2.3 Subcutaneous adrenaline (epinephrine)

Background

Adrenaline is a non-selective agonist of alpha and beta adrenoceptors that causes bronchodilation through stimulation of beta₂-adrenergic receptors leading to bronchial smooth muscle relaxation, while the alpha agonist effect reduces airway oedema through the stimulation of alpha receptors (111). Historically, subcutaneous or IM adrenaline was the mainstay of acute asthma treatment but current asthma guidelines do not recommend its use except in the context of concomitant anaphylaxis or angio-oedema (22, 111, 112). However, the administration of IM adrenaline in addition to nebulized selective beta₂ agonist for acute severe or life-threatening asthma remains part of many prehospital emergency guidelines and continues to be used by some clinicians (112, 113).

A recent systematic review and meta-analysis that evaluated the use of adrenaline administered by any route, compared to the administration of selective beta₂ agonist by any route in adults and children with acute asthma exacerbation (114). The review found limited evidence that there was no benefit from the use of adrenaline in addition to selective beta₂ agonist in acute asthma exacerbation (114). Subcutaneous adrenaline did not reduce treatment failure or lead to greater bronchodilation than nebulized selective beta₂ agonist alone.

WHO's *Pocket book of hospital care for children* currently advises the use of subcutaneous adrenaline in severe life-threatening asthma if inhaled or nebulized salbutamol is not available (1). This practice dates to a time when salbutamol was not widely available. There was need to evaluate the current evidence to support on the use of subcutaneous adrenaline in acute asthma exacerbations.

No recommendation:

WHO makes no recommendation on the use of subcutaneous adrenaline in the management of acute asthma.

Remarks

Adrenaline should be reserved for patients who present with symptoms and signs consistent with severe allergic reactions.

The use of IM adrenaline for anaphylaxis is described in WHO's *Pocket book of hospital care for children* (1) and the *Integrated management of adolescent and adult illness* (2).

Justification

The GDG agreed that the evidence was too indirect to make a recommendation as standard first-line therapy for acute asthma exacerbation was not used in either group. The GDG agreed that inhaled/nebulized SABA and systemic corticosteroids are the minimal standard of care for children with acute asthma exacerbations and should not be replaced by subcutaneous adrenaline. The use of subcutaneous or IM adrenaline should only be used in those who present with symptoms and signs consistent with severe allergic reactions.

Summary of evidence

The systematic review included one RCT comparing subcutaneous long-acting adrenaline (0.005 mg/kg) with nebulized SABA to nebulized SABA alone in 43 children with acute asthma exacerbation according to the study guidelines. Critical outcomes of interest included hospitalization, morbidity (improvement in symptoms) and adverse effects. No significant difference between hospitalization in the studied group was noted (RR: 0.38; 95% CI: 0.04–3.40) and no significant difference was found in the number of albuterol nebulizations needed between the two groups (MD: –0.20; 95% CI: –0.62 to 0.22). The study found no change between the groups in the assessment of heart rate and respiratory rate post intervention but did not specifically evaluate adverse effects (115).

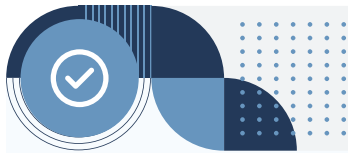
4.2.4 IV SABA in management of acute severe asthma exacerbation

Background

The current standard of care for the management of acute asthma exacerbations includes inhaled SABA and systemic corticosteroids (116). SABA is a β_2 -adrenoceptor agonist and binds to the β_2 receptor, thereby activating adenylyl cyclase which results in the conversion of adenosine triphosphate (ATP) to cyclic adenosine monophosphate (AMP). This promotes bronchodilation and effective symptom control in acute asthma exacerbation (22). The penetration of inhaled SABA, especially into the small conducting airways, may be reduced due to airway inflammation associated with severe acute asthma exacerbations, resulting in a reduced response. It is hypothesized that IV delivery of SABA to the receptors via the systemic circulation may improve clinical response (117).

A previous systematic review investigated the addition of IV SABA to inhaled SABA in 104 adults and children and included three RCTs (118). The outcome measures reported in the trials were heterogeneous and only one study reported statistically significant shortening in recovery time in the IV SABA (salbutamol) group ($P = 0.03$) (117). The IV SABA group was discharged 9.7 hours earlier than the inhaled SABA group ($P < 0.05$) (117). In the study conducted in the PICU, there was no benefit in length of admission in the IV SABA (terbutaline) group (MD: –12.95 hours; 95% CI: –38.74 to 12.84) (119). A subsequent observational study in 120 children found a 44% decrease in mechanical ventilation when IV SABA was used and the duration of mechanical ventilation was shorter in the group who received IV SABA prior to PICU admission (120).

Based on the limited current favourable evidence from single studies, the GDG judged the importance of defining the role of IV SABA in the management of acute asthma exacerbation as a priority problem.



Recommendation 4:

IV SABA should be used as a potential second-line therapy for severe paediatric asthma exacerbations in addition to standard first-line therapy with inhaled/nebulized medicines (SABA and ipratropium) and systemic corticosteroids in health care facilities.

Strong recommendation, high-certainty evidence.

Remarks

- IV SABA can be used as a potential second-line therapy, depending on the availability of other options. IV magnesium sulfate would be preferred as second-line therapy, if available (see Recommendation 5).
- IV SABA is most likely to be used in the PICU or settings where the infrastructure and monitoring already exist.
- Careful clinical monitoring is recommended to ensure the early identification and management of possible cardiovascular adverse effects, such as tachycardia, hypotension and arrhythmias.
- Where resources are available, it is advisable to have continuous ECG monitoring, and check serum electrolytes (hypokalaemia) and lactate.

Justification

Given the balance between the moderate desirable benefits, small undesirables and values, acceptability and feasibility, the GDG made a strong recommendation favour of the use of IV SABA in severe asthma exacerbations with poor response to standard first-line therapy including inhaled/nebulized SABA and systemic corticosteroids. Additional literature considered suggested that caregivers would value the moderate benefits of the intervention over the small risk of adverse effects (37). The GDG agreed that a strong recommendation would encourage the procurement of the IV formulations to ensure equitable access to IV SABA across settings.

Summary of evidence

This meta-analysis included four studies of 183 children aged 18 years and younger with acute severe asthma exacerbation as defined by the study criteria, mostly conducted in high-income settings except for one upper-middle-income country (Brazil) (75). Children were recruited from emergency departments, admission wards and PICUs and all children received and failed to improve with standard first-line therapy including at least three inhalations of SABA and systemic corticosteroids. The addition of IV SABA to standard first-line therapy was compared to standard first-line therapy alone or in addition to placebo. The dosing of SABA was varied, but typically included IV salbutamol 15–20 µg/kg over 10–20 minutes or terbutaline 10 mg/kg per min over 10–20 minutes followed by an infusion. The critical outcomes under investigation included the length of hospital stay and escalation of care (PICU admission or mechanical ventilation), morbidity (symptom improvement) and adverse events.

When compared to placebo, there was high-certainty evidence that duration in hospital stay was reduced by nearly 1 day in those who received IV SABA (MD: –23.45 hours; 95% CI: –36.09 to –10.81) (75, 117, 119, 121). Only one study reported on the need for mechanical ventilation and found no difference among groups (RR: 0.13; 95% CI: 0.01–2.42) (119). The need for oxygen was reported at varying timepoints in three studies (75, 117, 121). There was a reduction in oxygen dependence at 2 hours in the IV SABA group (RR: 0.27; 95% CI: 0.07–1.05), but the result was not significant (117). The duration of oxygen supplementation in days were less in the IV SABA group (MD: –1.73; 95% CI: –3.01 to –0.45) (75) and the need for supplemental oxygen at 12 hours was similar among groups, but rapid weaning occurred in those who received IV SABA and ipratropium bromide, rather than ipratropium bromide alone (RR: 0.26; 95% CI: 0.11–0.61) (121). One study showed clinical asthma severity scores improving after 2 hours in the IV SABA group, with a reduction in persistent moderate to severe asthma and persistent severe asthma (117). But another showed no difference between clinical scores when compared between groups at 24 hours (119).

Adverse effects were reported inconsistently. No clinically significant hyperglycaemia or hypokalaemia was observed (117, 121). There was a higher proportion of tremor at 2 hours in the IV SABA group ($P < 0.02$) (117). In one trial, tachycardia occurred in five (29.4%) children who received IV SABA (74). In those who received terbutaline, six (24%) had high troponin I values, and one had an arrhythmia but troponin I levels were not raised (121). Hypotension was documented in the IV terbutaline group (RR: 0.43), but all received IV fluid therapy, and none required inotropic support (121).

Benefits and desirable effects

The GDG judged the desirable effects to be moderate. There was high-certainty evidence that IV SABA reduced the length of hospital stay by 1 day (MD: 23.45 hours; 95% CI: 36.09–10.81). Single studies reported additional benefits such as a reduction in oxygen requirements (75, 121) and an improvement in clinical severity score in those with persistent moderate to severe asthma ($P < 0.002$) and persistent severe asthma ($P = 0.02$) (117).

Harms and undesirable effects

The reports on adverse effects associated with IV SABA use were heterogeneous with very low-certainty evidence from single studies reporting an increase in tremor (117), troponin I levels and hypotension not requiring inotropes (121). Additional evidence suggested that increased troponin I levels in children treated with IV SABA are common, but unlikely to be associated with harm (122, 123). The GDG therefore judged the undesirable effects to be small.

Other considerations

Preferences, values and acceptability: The GDG agreed that providers and caregivers would probably value the moderate benefits of the intervention over the small risk of adverse effects. This view is supported by a qualitative study of parents and caregivers of children presenting to hospital with a severe acute exacerbation of asthma in 12 countries (high- and middle-income) (37). Caregivers represented in this study highlighted length of hospital stays, ICU or high dependence unit admission, and treatment costs as the most important outcomes. The GDG agreed that this intervention would probably be acceptable to all key stakeholders.

Resources required: The additional cost incurred for the administration equipment and staff required to administer IV SABA as a slow bolus (20–30 minutes) or infusion may preclude its use from resource-limited settings and the cost saving associated with decreased oxygen requirements is unlikely to balance the expense. No studies assessed the cost or cost-effectiveness of IV SABA, but the GDG agreed that IV SABA would incur at least moderate costs and that cost-effectiveness would vary depending on the setting.

Equity: The GDG also considered how the significant upfront investment required to implement IV SABA availability may reduce equity and unfairly challenge settings with limited resources by the required use of these interventions. The GDG agreed that equitable access to IV SABA would be increased through advocacy and inclusion on the EMLC, but the overall effect would be a reduction in equity (12). The GDG agreed that a recommendation favouring IV SABA may also encourage greater availability.

Feasibility: Although the use of IV SABA achieves effective bronchodilation, non-selective stimulation of beta adrenoreceptors is associated with diverse adverse effects and the cardiovascular side-effects are of major concern (22). Although tachycardia, hypotension and arrhythmias were not reported in the meta-analysis, careful clinical monitoring is still recommended to ensure the early identification and management of the adverse effects should they occur. Admission to PICU or facilities with adequate monitoring is therefore recommended and it is likely that IV SABA will be used more frequently in settings where the infrastructure and monitoring already exist. Other monitoring advised: continuous ECG monitoring, serum electrolytes (hypokalaemia) and lactate.

Research gaps

- Investigating the costs of IV SABA administration in acute asthma exacerbations.
- Investigating the cost-effectiveness of implementing the use of IV SABA in acute asthma exacerbations.

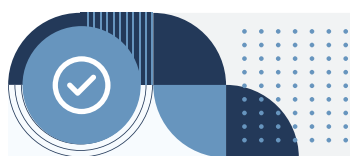
4.2.5 Preferred second- and third-line therapy for paediatric asthma exacerbations

Background

Current guidance on appropriate second-line therapy for acute asthma exacerbation in children who do not respond to standard first-line therapy or those presenting with life-threatening asthma is inconsistent. Previous reviews have suggested IV aminophylline, salbutamol, magnesium sulfate and adrenaline as various treatment options, but a recent systematic review was unable to make a firm recommendation (95, 124, 125). The review did however suggest that IV magnesium sulfate reduces hospital admission and length of stay, that hospital admissions are also reduced by the addition of ipratropium bromide to inhaled SABA and corticosteroids, and that IV aminophylline results in more vomiting when compared to IV SABA (95).

International guidelines currently suggest the use of subcutaneous adrenaline in children not responding to inhaled SABA and systemic corticosteroids (1) or an increase in the frequency of inhaled SABA with the addition of inhaled ipratropium bromide when available (11). Additional options also include a dose of IV magnesium sulfate or a terbutaline bolus given over 5 minutes, followed by a continuous infusion (22).

Due to the lack of clear evidence to support choice of second-line therapy for acute asthma exacerbation, the GDG judged the importance of defining the role of each IV therapy in the management of acute asthma exacerbation a priority problem.



Recommendation 5:

IV magnesium sulfate should be the preferred option for second-line therapy and IV SABA the preferred option for third-line therapy in severe paediatric asthma exacerbations not responding to standard first-line therapy.

Conditional recommendation, very low-certainty evidence.

Remarks

- A pragmatic stepwise approach to second-line therapies would be appropriate, and allow practitioners to choose the best possible option for their setting.
- In settings where neither preferred option is available, IV aminophylline may be considered as additional therapy in severe asthma exacerbations not responding to standard first-line therapy.

Justification

There was very low-certainty evidence in favour of magnesium sulfate to reduce hospitalization, reduce additional IV treatment and improve asthma severity score with no increase in undesirable effects. Moderate-certainty evidence is available in favour of IV SABA for its reduction of mechanical ventilation, a critical outcome, but at the cost of a small increase in arrhythmia. However, for IV aminophylline there was low-certainty evidence for the trivial benefit associated with length of hospital stay, a non-critical outcome, and moderate-certainty evidence in increase of undesirable adverse effects in the form of nausea and vomiting. The GDG agreed that due to contextual differences, resource limitations and a significant cost difference between therapies, a pragmatic approach to second-line therapies would be appropriate and allow practitioners to choose the best possible option for their setting.

The GDG agreed on a marginal benefit of IV magnesium sulfate, based on very low-certainty evidence, over IV SABA and IV aminophylline on balance of risk and benefit. However, a stepwise approach to IV therapy from magnesium sulfate to SABA and then aminophylline was advised. The GDG therefore recommended the use of IV magnesium sulfate as the preferred option for second-line therapy over IV SABA (second choice) and over aminophylline (third choice) in patients with acute severe asthma exacerbations not responding to standard first-line therapy.

Summary of evidence

The meta-analysis investigated studies of comparative interventions to identify second-line therapeutic options for acute severe asthma exacerbations. The available studies were analysed in three comparative groups, including IV aminophylline versus other treatment, IV SABA versus other treatment and IV magnesium sulfate versus other treatment.

IV aminophylline versus other treatment

A total of six RCTs of 410 children aged 18 years and younger were included. Three trials were conducted in low-middle-income and upper-middle-income countries (Venezuela (Bolivarian Republic of), India) (126–128). Children were recruited from the PICU in two studies and the emergency department, in which all had critical or severe asthma in three studies and moderate to severe asthma in one study (126–131). All children received and failed to improve with standard acute asthma therapy, including at least three inhalations of SABA and systemic corticosteroids. IV aminophylline (5 mg/kg) was typically administered as a slow bolus over 20 minutes followed by an infusion (1 mg/kg per min) and compared to a slow bolus of magnesium sulfate (50 mg/kg), a slow bolus IV SABA (15 µg/kg salbutamol or 120 µg/kg terbutaline) and a slow bolus of IV ketamine (0.5 mg/kg) followed by an infusion (0.6 mg/kg per hour) (126–131).

A significantly higher hospitalization rate was reported in the IV aminophylline group compared to IV magnesium sulfate (RR: 1.52; 95% CI: 1.26–1.84) (128). There was low-certainty evidence that IV aminophylline decreased length of hospital stay in a pooled comparison to IV SABA and ketamine (MD: –12.99 hours; 95% CI: –33.13 to 7.15) (127, 130). The rate of PICU admission was no different between the IV aminophylline and IV magnesium sulfate (RR: 1.72; 95% CI: 0.53–5.61) groups, and the duration of PICU admission in hours was also no different in the IV aminophylline and IV SABA groups (MD: –12.00; 95% CI: –58.31 to 34.31) (128, 129). There was no difference in asthma severity scores when aminophylline was compared to any other treatment (MD: 0.97; 95% CI: –0.91 to 2.85) (127–130). The need for mechanical ventilation was no different between the IV aminophylline and IV SABA groups (RR: 0.35; 95% CI: 0.03–3.54) (130). The aminophylline group suffered more nausea (RR: 2.80; 95% CI: 1.02–7.70) and vomiting (RR: 1.74; 95% CI: 1.10–2.75) compared with any other treatment group (126, 127, 129, 131). Overall, there was low-certainty evidence of trivial benefits associated with IV aminophylline for length of hospital stay, which is a non-critical outcome, but moderate-certainty evidence that IV aminophylline had a moderate increase in undesirable adverse effects in the form of nausea and vomiting.

IV SABA versus other treatment

Two RCTs including 92 children aged 18 years and younger were included in the review. One study was conducted in the Netherlands (Kingdom of the) and the other in Brazil (75, 132). One study recruited children who required hospitalization after failing to respond to standard acute asthma therapy including at least three doses of inhaled SABA and systemic steroids, and one recruited those admitted to the PICU with severe asthma (75, 132). IV salbutamol (20 µg/kg) was compared to magnesium sulfate (50 mg/kg) and placebo (75, 132).

The IV SABA group, in a pooled comparison to IV magnesium sulfate or placebo, had an average shorter hospital stay of 10 hours (MD: –10.12; 95% CI: –45.74 to 25.49) and improved asthma severity scores (MD: 1.00; 95% CI: 0.61–1.39), but there was no difference in the need for mechanical ventilation (RR: 0.25; 95% CI: 0.03–2.21) (75, 132). There was no difference in the occurrence of adverse effects (RR: 1.13; 95% CI: 0.25–5.08, including arrhythmias (RR: 1.34; 95% CI: 0.10–17.36) between groups (75, 132). Overall, there was moderate-certainty evidence that IV SABA reduced mechanical ventilation, a critical outcome, but at the cost of a small increase in arrhythmia.

IV magnesium sulfate versus other treatment

One RCT recruited 28 children aged 18 years and younger admitted to hospital after failure to respond to standard acute asthma therapy, including at least three doses of inhaled SABA and systemic steroids. The study was conducted in Thailand, and included children with severe asthma exacerbation as defined by the study criteria. A slow bolus of IV magnesium sulfate (50 mg/kg) was compared to three doses of inhaled magnesium sulfate (133). The primary outcomes investigated included the length of hospital stay

and escalation of care (PICU admission or mechanical ventilation), morbidity (symptom improvement) and adverse events. The length of hospital stay (MD: 12.96 hours; 95% CI: –25.16 to 51.08) and asthma severity scores (MD: 0.30; 95% CI: 0.36–0.96) did not differ between groups receiving IV or nebulized magnesium sulfate. Overall, there was very low-certainty evidence that magnesium sulfate reduced hospitalization, reduced additional IV treatment (RR: 7.88) and improved asthma severity score (MD: 0.97) with no increase in undesirable effects.

Benefits and desirable effects

The GDG agreed that the studies were underpowered to determine critical outcomes, including clinical improvement and the need for mechanical ventilation, for which a very large number of participants is required. They also agreed that using the length of hospital stay, which favoured aminophylline (MD: 12.99 hours) as a proxy outcome for clinical improvement, is problematic, since it may be influenced by various aspects of health care delivery (timing of ward rounds, availability of a pharmacist) and is not solely dependent on the underlying disease process. The GDG judged the desirable effects of aminophylline over other second-line therapies to be trivial. The GDG agreed that very low-certainty evidence favoured magnesium sulfate for hospitalization rate (RR: 1.52), less IV treatment (RR: 7.88) and improvement in asthma severity scores (MD: 0.97) and agreed that the superiority of IV magnesium sulfate was consistent across studies. The GDG also considered the moderate-certainty evidence that IV SABA was favourable in preventing the need for mechanical ventilation (RRR: 4.9%), but at the cost of an increase in arrhythmia (RRI: 0.8%, moderate-certainty evidence).

Harms and undesirable effects

The GDG judged the undesirable effects of IV aminophylline compared to other agents as moderate, based on the increase in nausea (RRI: 21.9%, low-certainty evidence) and vomiting (RRI: 17.3%, moderate-certainty evidence). The GDG also considered the moderate-certainty evidence that SABA results in a small increase in arrhythmia (RRI: 0.8%). Compared to IV aminophylline and SABA, magnesium sulfate, for which very few non-severe adverse effects have been reported, has a superior safety profile.

Other considerations

Preferences, values and acceptability: The GDG agreed that a pragmatic approach to the implementation of second- and third-line therapies is required and that, taking contextual differences into account, these interventions would be valued and acceptable to all caregivers, health care providers and stakeholders involved.

Resources required: Although no studies with head-to-head cost and cost-effectiveness comparisons were included in the meta-analysis, there will be a cost difference between different therapies, especially those for which additional equipment (infusion pumps) and increased monitoring is required, such as IV SABA. The costs would vary among different settings, and it would likely limit the use of IV SABA. The GDG judged the substantial costs associated with the additional equipment, monitoring and training required for IV SABA administration less favourable than the minimal cost and monitoring required for magnesium sulfate administration. Therefore, the GDG agreed that the stepwise approach would be valued, acceptable and equitable to all the stakeholders.

Equity: The GDG also considered that the wide availability of aminophylline may perpetuate its use as a second-line therapy when other options are not available.

Feasibility: The GDG agreed a marginal benefit of IV magnesium sulfate, based on very low-certainty evidence, over IV SABA and IV aminophylline on balance of risk and benefit and a stepwise approach to IV therapy from magnesium sulfate to SABA and then aminophylline is advised. In addition, IV magnesium sulfate is widely available for obstetric use across settings which is most likely to favour its use.

Implementation considerations

- A pragmatic stepwise approach to second-line therapies would be appropriate and allow practitioners to choose the best possible option for their setting.
- The burden of additional equipment and monitoring required for the IV administration of second-line asthma therapies, which might be a challenge and may limit its use in resource-limited settings.
- The additional cost of IV SABA is most likely to limit its use in resource-limited settings, hence the need for advocacy for its use in areas where the pre-existing infrastructure allows.
- Since IV aminophylline is more widely available, it may have a role as second-line therapy, in the absence of alternatives in the stepwise approach.

Research gaps

- Investigating the benefit of IM magnesium sulfate administration as a rescue therapy in children with acute severe asthma exacerbation.
- Investigating the use of inhaled magnesium sulfate as a rescue therapy in children with acute severe asthma exacerbations.
- Is peak flow the optimal measurement in the evaluation of moderate to severe asthma exacerbations?

4.2.6 Combination inhaler MART for the long-term management of asthma

Background

The long-term management and control of asthma symptoms, particularly in resource-limited settings, is largely inadequate, with approximately one in 20 children and adolescents experiencing severe asthma symptoms (21). Uncontrolled asthma symptoms have a significant impact on daily life, causing school absence, impacting negatively on education and social opportunities (7). Asthma exacerbations require medical review, escalation of treatment and possibly hospitalization. Without timely and effective treatment, severe exacerbations can lead to death.

For patients experiencing frequent symptoms requiring relief with SABA, standard practice is to introduce regular treatment with daily ICS (134). Regular treatment with ICS is well established to reduce asthma-related deaths, hospitalizations and exacerbations (135-137). According to the Global Asthma Network Phase I study, the proportion of children worldwide with severe symptoms taking SABA alone, without additional ICS, was 44.8%, with significantly higher rates in low- and lower-middle-income countries (6). In children using SABA monotherapy, the use of three or more canisters per year is associated with a twofold risk of asthma-related hospitalization or emergency department visit, with each additional canister used per year associated with an 8–14% increased risk of acute asthma exacerbations (138).

Adherence to daily ICS treatment is considered a challenge, and over the past decade there has been increasing evidence to support a new model of long-term management, using a combination inhaler containing both ICS (such as budesonide) and a rapid-onset long-acting beta₂ agonist (LABA) (such as formoterol). Rapid-onset LABA, which acts as quickly as SABA (such as salbutamol or terbutaline), can be used in a combination inhaler with ICS and taken as needed to relieve asthma symptoms, and also for regular maintenance therapy in those with more persistent symptoms (22). This has the advantage of simplifying an inhaler regime and ensuring that LABA therapy is not taken without ICS.

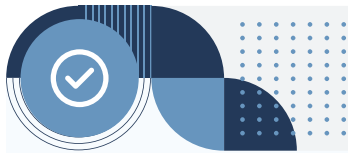
Systematic reviews assessing the use of combination inhalers as reliever or MART have included patients aged 12 years and older, and report favourable reductions in exacerbations and decreased exposure to inhaled and systemic steroids (139-141).

A systematic review of combination budesonide/formoterol MART versus current best practice (including inhaled steroid maintenance) conducted in 2013 reported a reduction in exacerbations requiring oral steroids (OR: 0.83; 95% CI: 0.70–0.98) and requiring hospitalization (OR: 0.81; 95% CI: 0.45–1.44) for adults and adolescents (aged 12 years and older) receiving MART. Of the 224 children (aged 4–11 years) included in one trial, only nine on maintenance combination inhaler therapy had an exacerbation requiring medical intervention compared to 21 on ICS (OR: 0.33; 95% CI: 0.15–0.77) (141). A later systematic review of MART in patients with persistent asthma did not identify any additional paediatric studies, concluding that evidence for children is limited. However, for those aged 12 years and older ($n = 22\,749$, mean age 42 years), the data showed that the use of MART compared with ICS as maintenance therapy (with or without LABA) was associated with a lower risk of asthma exacerbations (140). Subgroup analysis of those aged 12–18 years has not been published.

A systematic review conducted in 2021 assessed the use of combination fixed-dose beta-agonist and steroid inhaler as required for adults and children with mild asthma and included five studies, with two trials enrolling patients aged 12 years and older. Use of the combination inhaler compared to SABA (salbutamol or terbutaline) alone resulted in reduced exacerbations (OR: 0.45; 95% CI: 0.34–0.60), hospital admissions or unscheduled health care visits (OR: 0.35; 95% CI: 0.20–0.60), and an overall reduction in exposure to systemic corticosteroids (MD: 9.9 mg prednisolone lower). Using a combination inhaler as needed compared to regular ICS use appears to reduce asthma-related hospital admissions and unscheduled health care visits (OR: 0.63; 95% CI: 0.44–0.91) and reduce average exposure to ICS. There were no data included for children aged younger than 12 years (139).

The use of a combination ICS-formoterol inhaler as reliever therapy alone or MART has been recommended by GINA as the preferred approach in the management of chronic asthma in adults and adolescents since 2019 (142). For children aged 6–11 years, current GINA recommendations are to consider ICS-formoterol MART as an alternative to medium dose ICS or low-dose ICS-LABA plus SABA reliever for those with asthma symptoms on most days (GINA treatment step 3) (22). A combination inhaler approach is not recommended by GINA for children aged ≤ 5 years due to insufficient data on efficacy and safety (22).

The GDG considered this evolving area of clinical practice to be a priority topic for consideration and review of the latest evidence.



Recommendation 6:

Combination inhaler (steroid plus fast-onset LABA) MART should be considered for children with persistent asthma symptoms requiring regular ICS treatment.

Conditional recommendation, moderate-certainty evidence.

Remark:

- Although evidence was considered for children aged 4–11 years, current formulations are licensed for children aged ≥ 6 years; hence, the recommendation is restricted to children aged 6–11 years.
- Current evidence supports the use of combination inhaler (steroid plus fast-onset LABA) MART for children with persistent asthma symptoms who are already taking or planning to start regular ICS therapy.
- The use of a combination inhaler results in a potentially simpler regime, which may be more easily followed by both patients and prescribers (143).
- Children with infrequent symptoms should use SABA as needed, considering the concomitant intermittent use of ICS.

Justification

Evidence from a single well-designed RCT reported decreased asthma exacerbation rates overall and decreased asthma exacerbation rates requiring medical intervention among children aged 4–11 years using a combination budesonide/formoterol inhaler for MART. Safety data from this RCT and additional data relating to inhaled corticosteroids/long-acting beta₂ agonist (ICS/LABA) combination inhaler was favourable. Cost-effectiveness studies have also reported favourable outcomes for budesonide/formoterol MART, compared to other treatment regimes.

Although there are several publications on combination inhaler MART that included participants aged 12 years and older, it was not possible to extract the data for those aged 12–18 years for review by the GDG. The GDG therefore preferred to restrict their recommendation to the age group for which data were available and for which current formulations are licensed.

The GDG agreed that the evidence probably favoured the intervention, in keeping with the findings from the greater body of evidence from adult literature. However, given the relative lack of evidence in children, the GDG chose to make a conditional, rather than strong recommendation. The GDG reflected that the likely uptake of a combination inhaler MART approach would depend on the relative costs of the combination inhaler, compared to individual SABA and ICS inhalers.

Summary of evidence

Only one study was identified which assessed the use of combination inhaler MART in children (144). The data were extracted, as part of a prospectively planned analysis, from a study that recruited patients aged 4–80 years. Participants were recruited at 41 centres across five upper-middle-income and seven high-income countries. The three-arm double-blind RCT included 341 children, aged 4–11 years, with at least a 6-month history of asthma, and at least one clinically important exacerbation in the preceding 12 months. All children had been treated with ICS for at least 3 months.

A MART regimen (budesonide/formoterol 80/4.5 µg once daily maintenance plus additional inhalations for symptom relief) was compared to budesonide/formoterol 80/4.5 µg once daily for maintenance (fixed combination) or higher-dose budesonide 320 µg once daily (fixed-dose budesonide) ($n = 106$). Blinded as-needed SABA medication (terbutaline 0.4 mg) was provided in both fixed-dose groups. Participants were followed up for 12 months.

The critical outcomes identified by the GDG reported for this study included overall exacerbation rates, exacerbations requiring medical intervention (hospitalization or emergency department treatment, treatment with oral steroids, an increase in ICS and/or additional treatment), adverse events, discontinuation of treatment and daily inhaled steroid dose. Changes in lung function were considered less important by the GDG. The methodological quality of this study was assessed using the Cochrane risk-of-bias tool for randomized trials (145) and the certainty of the evidence was determined through the GRADEpro guideline development tool (146).

The overall exacerbation rates were 14% (17 out of 118 patients) on the MART regimen compared to 38% (44 out of 117 patients) on the fixed-dose combination (RR: 0.38; 95% CI: 0.23–0.63) and 26% (28 out of 106 patients) on fixed-dose budesonide (RR: 0.55; 95% CI: 0.32–0.94). The rates of exacerbations requiring medical intervention were 8% (10 out of 118 patients) for patients on MART compared to 31% (36 out of 117 patients) on the fixed-dose combination (RR: 0.28; 95% CI: 0.14–0.53) and 20% (21 out of 106 patients) on fixed-dose budesonide (RR: 0.43; 95% CI: 0.21–0.87).

The mean daily ICS dose was 126 µg for those using the MART regime, compared to 80 µg for the fixed-dose combination and 320 µg for fixed-dose budesonide.

PEF measurements were higher in those receiving MART (mean morning PEF: 255 L/min) compared to fixed-dose budesonide (mean morning PEF: 238 L/min; $P = 0.002$) and fixed-dose combination (mean morning PEF: 242 L/min; $P = 0.22$). FEV₁ measurements were similar in all groups.

Serious adverse events were uncommon and included asthma exacerbations, fractures and pneumonia. The rate of serious adverse events was lower among those receiving MART (2 out of 118 patients), compared to fixed-dose combination (16 out of 117 patients) (RR: 0.11; 95% CI: 0.02–0.48) and fixed-dose budesonide (5 out of 106 patients) (RR: 0.35; 95% CI: 0.07–1.83). Class-related effects, such as tremor, dysphonia and tachycardia were rare and evenly distributed between the three groups.

There was no difference in discontinuation of treatment between those receiving MART (2 out of 118 patients) and those in the two other treatment arms (2 out of 223 patients) (RR: 1.89; 95% CI: 0.27–13.25).

Benefits and desirable effects

The GDG highlighted that the intervention under consideration was the use of a combination ICS/LABA inhaler to be used for both daily maintenance therapy, and for additional reliever therapy as needed. Trials assessing the use of a combination ICS/LABA inhaler for as-needed therapy only, in those with milder asthma, were not included in the review of evidence.

Overall exacerbation rates and exacerbations requiring medical intervention were reduced in participants receiving MART, compared to those using a fixed-dose combination inhalers or a fixed-dose budesonide inhaler. The use of a composite outcome “exacerbations requiring medical intervention” meant that it was not possible to determine the specific effect on rates of hospitalization or emergency health care use. The impact on lung function was minimal, with slightly higher PEF measurements recorded for the MART group that are unlikely to be of clinical significance, and no difference in FEV₁ between the groups (147).

Certainty of evidence for all outcomes was downgraded to moderate, on the basis that the single RCT was sponsored by a pharmaceutical company. The GDG judged the desirable effects to be moderate.

Harms and undesirable effects

In this single RCT, the number of reported adverse events was small and favoured the use of MART. The daily dose of ICS was fairly comparable between those that received MART and those using the fixed-dose combination inhaler and much higher in those using the fixed-dose budesonide inhaler.

The GDG also considered additional safety data in children receiving ICS/LABA in a fixed-dose combination. Budesonide/formoterol (delivered by both metered-dose and dry-powder inhalers) was well tolerated, with individual mild adverse events such as dysphonia, oral candidiasis, muscle cramps, palpitations, tremors and tachycardia occurring in <0.2% of the overall population (148).

The GDG judged the undesirable effects to be small, based on moderate-certainty evidence, noting that the follow-up period of 12 months would be too short to determine long-term adverse effects.

Other considerations

Preferences, values and acceptability: The GDG agreed that the critical outcomes under consideration, such as reduction of exacerbations, particularly those requiring medical intervention, such as hospitalization and emergency department treatment, would be valued by children with asthma, their families and health care providers. The GDG agreed that combination inhaler MART could represent an attractive alternative for patients, providers and health systems if it can be provided at a similar cost to conventional asthma treatment.

Resources required: A cost-effectiveness study in a paediatric population in China reported decreased outpatient service costs with budesonide/formoterol MART and comparable drug costs to alternative treatment regimes. MART reduced the total cost by approximately CNY 10 516 per year with a 0.12 quality-adjusted life-year increase (149). A cost-utility study, comparing three alternative regimes for the management of moderate asthma in children aged 6–11 years from Colombia, found that budesonide/formoterol MART was more cost-effective than fixed combination budesonide/formoterol and fixed-dose budesonide (150). GDG members agreed that the cost and cost-effectiveness of implementing a MART approach would vary between different countries, depending on many factors.

The availability of combination inhalers is limited in resource-constrained settings: WHO's 2021 NCD Country Capacity Survey reported that combination budesonide-formoterol inhalers were generally available in public health sector primary care facilities in 7% of low-income countries, 26% of lower-middle-income countries and 42% of upper-middle-income countries (151). However, availability could be facilitated if included on the EMLc. Budesonide/formoterol dry-powder inhaler (100/6 µg/dose; 200/6 µg/dose) has been on the WHO Essential Medicine List for adults since 2017 as single-inhaler therapy for asthma, indicated for regular maintenance therapy, to control the disease in patients who have failed first-line therapy, but not recommended for reliever or rescue therapy (152).

Equity: GDG members discussed concerns regarding equity due to the likely higher costs of the inhalers, compared to conventional SABA and ICS, and the need to familiarize and train health care workers. The GDG discussed that it would take time and effort to make combination inhalers widely available and affordable.

Feasibility: Feasibility would depend on ability of children to use the inhaler device effectively. In the only trial reviewed, the medicine was delivered using a turbuhaler, which is a type of dry-powder inhaler. The device is breath activated, so the dry-powder medication is automatically released when the child inhales. The majority of MART trials in the adult and adolescent population have been conducted using a dry-powder inhaler. Dry-powder inhalers may be considered for use by children aged as young as 3 years. Metered-dose inhalers containing this combination are also available. The GDG highlighted that spacer devices should be available to promote the effective use of MDIs in younger children. Where commercial spacer devices are not available, a 500 mL bottle spacer is an effective alternative, but should be constructed following the methods that have been demonstrated as effective (59).

Stringent and national regulatory authority approvals for combination inhalers, including specific age groups and indications, will also affect the uptake of MART. In 2017, Symbicort (budesonide/formoterol fumarate dihydrate, NDA: 021929) was approved by the United States Food and Drug Administration for maintenance treatment of asthma in children aged 6–12 years, having previously been approved for patients aged 12 years and older in 2006 (153).

Implementation considerations

- The importance of ensuring that all asthma medicines, whether for reliever or maintenance therapy, are delivered by an appropriate inhaler device, considering the child's age and preference.
- Where treatment is given using an MDI, it is essential that spacer devices are available to enhance delivery of the medicine into the lungs.
- In the absence of commercial spacer devices, a 500 mL bottle spacer is an effective alternative but should be constructed following the methods that have been demonstrated as effective (59).
- For both metered-dose and dry-powder inhalers, careful explanation and regular review of inhaler technique are vital.
- Successful adoption of a new approach to asthma management would require comprehensive training of health care providers to ensure competence to prescribe appropriately and educate patients and their families accordingly.
- Education on inhaler use, specific to the chosen delivery device, is an essential part of asthma management.

Research gaps

There is a need for more data on the clinical and cost-effectiveness of ICS-LABA combination inhalers across the spectrum of asthma severity in children and adolescents, particularly in resource-limited settings.

Exploration of different service-delivery models for the diagnosis and management of asthma in primary health care in resource-limited settings.

5. Management of bronchiolitis in infants and young children



Bronchiolitis is an acute viral infection of the lower respiratory tract that typically affects infants and children aged less than 2 years, and is the leading cause of hospitalization in children globally (154). It is primarily caused by the respiratory syncytial virus (RSV), but other viruses including adenovirus, human metapneumovirus, influenza and parainfluenza contribute significantly (154, 155). Children with viral bronchiolitis are at increased risk of invasive bacterial infections, including pneumonia, and have a higher risk of developing asthma and recurrent wheezing in later childhood (156-158). Severe disease is more common among preterm babies, children younger than 6 months and children with comorbidities, specifically, the immunocompromised and children with congenital heart disease, neuromuscular disorders and other chronic neurological conditions, chronic lung disease, and asthma or reactive airway disease (159-161). Estimates suggest that RSV disease results in more than 30 million cases of lower respiratory tract infection in children aged less than 5 years annually, with 3.2 million hospitalizations and 200 000 deaths worldwide each year (162). Acute bronchiolitis is also associated with a high economic burden from the utilization of health care resources, with the highest costs attributable to severe bronchiolitis in children with comorbidities and prematurity (163).

5.1 Clinical presentation

Acute bronchiolitis typically presents with acute onset of upper respiratory prodrome of rhinitis or congestion, low-grade fever and cough progressing to the symptoms of increasing respiratory distress (tachypnoea and increased work of breathing) and wheezing (155, 164). Common signs include tachycardia, tachypnoea, hyperinflated chest, hyperresonant sounds on chest percussion, wheezing and crackles. In children with severe disease, there may be grunting, marked retractions, cyanosis or lethargy. Apnoea is common in preterm babies and those aged less than 2 months. The severity of bronchiolitis can vary from mild symptoms to acute respiratory failure, but most cases are self-limiting and respond to symptomatic care at home. Complications among the hospitalized may include severe respiratory distress or failure, pneumothorax, myocardial injury, neutropenia, sepsis, shock and neurological complications (165, 166).

5.2 Diagnosis and management

The diagnosis and assessment of acute bronchiolitis is based on history and clinical examination. Laboratory investigations, chest radiographs and virological studies are not routinely indicated (13, 164). The majority of children can be managed at home with supportive care on an ambulatory basis. Home care includes parental education about adequate feeding, nasal saline drops and the gentle suctioning of nostrils to relieve nasal block, and early care-seeking in case of any worsening symptoms (13, 164). The current practice for those hospitalized is supportive care, including humidified oxygen therapy, adequate hydration and feeding, and close monitoring, with no specific interventions of proven benefit despite wide variability in the acute bronchiolitis care (13, 164, 167). Children with severe respiratory distress may require invasive mechanical ventilation, which is associated with various complications (ventilator-induced lung injury, infection, airway trauma, vocal cord dysfunction and need for prolonged sedation) and significant financial costs (168). However, various non-invasive modes of oxygen delivery are increasingly being used, such as HFNC, CPAP, non-invasive positive pressure ventilation and bilevel positive airway pressure, which may obviate the need for invasive mechanical ventilation and the complications related to it (169-173).

5.3 Scope and target audience

This guideline reviews the current evidence on selected topics related to the diagnosis and management of bronchiolitis in children aged less than 2 years. It does not encompass all aspects of the management of bronchiolitis, but discusses the priority clinical aspects for primary and secondary care levels, focusing on controversial areas of clinical diagnosis and treatment, and addresses the following topic areas in detail:

- clinical signs and symptoms for the diagnosis of bronchiolitis;
- clinical indicators for admission;
- the role of X-ray imaging in bronchiolitis;
- oxygen therapy and delivery in the treatment of bronchiolitis; and
- the use of nebulized normal saline in the treatment of bronchiolitis.

The prioritized PICO questions discussed in section 6 are outlined in Annex 2.

The target audience for these recommendations are policy-makers and health care professionals working in diverse settings and with variable availability of resources, in particular those in LMICs.

6. Recommendations for the management of bronchiolitis in infants and young children



6.1 Clinical signs and symptoms for the diagnosis of bronchiolitis



Guiding principle for using clinical symptoms and signs in supporting a probable diagnosis of bronchiolitis in an infant or child

Good practice statement 1: In infants and children aged less than 2 years presenting with cough and difficulty in breathing, a comprehensive history and assessment of clinical signs and symptoms should be done. A diagnosis of bronchiolitis can be suspected if they have a history of a prodrome of an upper respiratory tract infection with rhinitis or coryza and cough, followed by respiratory distress with a low-grade fever, and any presence of tachypnoea (fast breathing), chest indrawing and diffuse wheeze or crackles on auscultation.

Remarks

- Assessment of clinical signs should include the IMCI (3) general danger signs,⁴ cyanosis, respiratory rate, looking for chest indrawing, listening for a wheeze or crackles, and measuring oxygen saturation.
- Infants with viral bronchiolitis are usually not very ill looking, and many have a prolonged expiratory phase of breathing associated with wheeze.
- During an RSV outbreak, a higher index of suspicion should be maintained in the differential diagnosis for infants presenting with respiratory symptoms.
- A comprehensive history should also elicit presence of any risk factors and comorbidities, such as prematurity, children who are immunocompromized, and children with asthma or reactive airway disease, chronic lung disease, chronic neurological conditions or congenital heart conditions, and breastfeeding status.

6.1.1 Summary of evidence

The systematic review identified six observational studies from five countries (Australia, India, Iran (Islamic Republic of), Saudi Arabia and Viet Nam) with a total sample size of 1396 participants (174-179). Outcomes of interest were correct diagnosis, sensitivity and specificity, and the appropriate use of antibiotics. None of the studies looked at the diagnostic accuracy of the signs and symptoms or whether they were associated with the appropriate use of antibiotics. All studies were assessed as having high risk of bias owing to study design, some were retrospective with missing data, or had a low sample size of less than 100, or had reporting concerns.

4 The IMCI general danger signs are: inability to drink or breastfeed, vomiting everything, experiencing convulsions and being lethargic or unconscious.

Table 2. Prevalence of signs and symptoms in infants clinically diagnosed with viral bronchiolitis in the included studies

Sign or symptom	Studies (N=6)	Sample size	Overall prevalence (clinical) (% 95% CI)	RSV positive samples	RSV prevalence (% 95% CI)
Cough	6	1 396	64 (34–94)	986	96 (93–99)
Difficulty in breathing	6	986	56 (20–92)	826	81 (71–91)
Tachypnoea	5	1 164	52 (8–96)	811	78 (67–90)
Nasal discharge	5	1 223	52 (8–96)	858	76 (56–96)
Wheeze	5	1 164	51 (6–95)	811	74 (49–99)
Fever	5	1 164	48 (22–73)	811	71 (62–80)
Auscultatory sounds	5	1 164	44 (3–92)	811	65 (33–98)
GIT	3	496	22 (3–40)	295	39 (33–44)
Cyanosis	2	467	11 (9–14)	286	21 (16–26)
Hypoxia	2	234	5 (3–7)	286	5 (3–9)

CI: confidence interval; GIT: gastrointestinal tract; RSV: respiratory syncytial virus.

The most common symptoms and signs were cough (64%), difficulty in breathing (56%), tachypnoea (52%), nasal discharge (52%), wheeze (51%), fever (48%) and auscultatory sounds (44%) (Table 2). The GDG observed that these signs were identified in cases suspected to have a clinical diagnosis of bronchiolitis that was based on the very same signs and symptoms on which the diagnosis was made.

6.1.2 Justification

A systematic review identified only six observational studies of *suspected* cases of bronchiolitis, where the diagnosis was not confirmed. The major factors that were likely to be predictive of bronchiolitis were cough, nasal discharge, difficulty in breathing, fever, tachypnoea, wheeze and auscultatory sounds, which align with current clinical practice. However, the GDG noted that these signs were identified in cases suspected to have a clinical diagnosis of bronchiolitis, which is based on the very same signs and symptoms used for the diagnosis. In this respect, the GDG determined that, although a recommendation was not feasible based on the available evidence, a good practice statement could be made in line with the *WHO handbook for guideline development* (15), which would help clinicians make a diagnosis. This statement would guide clinicians in diagnosing bronchiolitis, since the reviewed signs and symptoms align with standard clinical practice and the counterfactual would not make sense.

6.1.3 Implementation considerations

Implementation will require clinicians to take into consideration the following.

- There is usually an overlap in the clinical presentation of viral bronchiolitis and bacterial pneumonia, and sometimes both exist or bacterial pneumonia may follow a viral upper respiratory infection.
- Differential diagnoses to be considered include viral or bacterial pneumonia, and other serious conditions, such as cardiac failure, gastro-oesophageal reflux, foreign body, tuberculosis, etc.
- Assessment of clinical signs should include IMCI general danger signs,⁵ cyanosis, respiratory rate, looking for chest indrawing, listening for a wheeze or crackles, and measuring oxygen saturation.
- A child who has IMCI general danger signs, who looks very unwell, has a high fever (temperature >38°C) or a fever persisting for 5 days or more, or is very hypoxic should not be assumed to have viral bronchiolitis alone.

6.1.4 Research gaps

- There is a need for a properly powered study that looks at the diagnostic accuracy of the signs and symptoms of bronchiolitis.
- There is a need to determine the signs and symptoms of bronchiolitis that are associated with appropriate use of antibiotics.

⁵ The IMCI general danger signs are: inability to drink or breastfeed, vomiting everything, experiencing convulsions and being lethargic or unconscious.

6.2 Clinical criteria for hospital admissions of patients with bronchiolitis



Guiding principle for assessing infants and children with suspected bronchiolitis

Good practice statement 2: In infants and children aged less than 2 years presenting with suspected bronchiolitis, clinicians should perform a comprehensive assessment for the presence of risk factors and signs of severe disease.

Indications for admission include: signs of severe disease, which include IMCI general danger signs (inability to breastfeed or drink, vomiting everything, convulsions and lethargy or unconsciousness), high temperature of $\geq 38^{\circ}\text{C}$, central cyanosis or $\text{SpO}_2 < 90\%$, severe respiratory distress, apnoea or history of apnoea, and presence of risk factors for severe disease, which include comorbidities, a prematurity and age < 3 months.

Remarks

- The indications for admission found in the review are mostly similar to the current WHO guideline recommendations for the admission of severe pneumonia (1).
- Clinicians should avoid unwarranted admissions that may lead to exposure to nosocomial infections and avoidable hospital care costs.
- Local context should be taken into consideration, particularly the availability of follow-up or its accessibility, including travelling distances for patients, for example.
- The temperature threshold of $\geq 38^{\circ}\text{C}$ in cases of suspected bronchiolitis is significant and might be justification for admission or close observation.
- Comorbidities to consider in the history include: congenital cardiac lesions, chronic lung disease, cerebral palsy and neurological conditions.

6.2.1 Justification

This guiding principle was included because an assessment of clinical signs and symptoms when making clinical management decisions on bronchiolitis would never *not* be performed. Clinicians need to perform a comprehensive assessment of the severity of signs and symptoms to decide which children with bronchiolitis need hospitalization. These signs are commonly used to assess sick children for severe disease, particularly respiratory involvement, and to determine when escalation of care is warranted.

6.2.2 Summary of evidence

The systematic review identified eight studies that were conducted in outpatients, emergency departments or wards of tertiary hospitals in Asia and Europe with a total of 473 607 participants, 95% of which were from two very large cohort studies (180-187). The critical outcomes measured were hospital admission, length of stay in hospital, ICU admission and mortality. Six of these studies had a high risk of bias, mainly driven by their retrospective design, missing data or misclassification or measurement biases made in ascertaining diagnoses of bronchiolitis retrospectively. Table 3 shows the indications for admission in children with bronchiolitis aged less than 2 years in these studies.

Table 3. Indications for admission in children with bronchiolitis aged less than 2 years

Sign or symptom	Adjusted odds ratio (95% CI)	P value	Sample size	Study
<i>Respiratory system</i>				
Respiratory rate	1.07 (1.03–1.11)	0.001	260	Amat (180)
Tachypnoea	4.68 (2.50–8.76)	<0.001		
Chest retractions	2.96 (1.53–5.69)	0.001	240	Mahajan (183)
SpO ₂ <92%	3.68 (1.67–8.13)	0.001		
Oxygen saturation	0.67 (0.54–0.85)	0.001	260	Amat (180)
<i>Nervous system</i>				
Fever	2.39 (1.7–3.33)	0.001	4 793	Masarweh (184)
	2.44 (1.06–5.62)	0.036		
GCS <15	7.75 (1.47–40.9)	0.016	240	Mahajan (183)

GCS: Glasgow Coma Scale; SpO₂: peripheral capillary oxygen saturation.

A higher respiratory rate resulted in increased admissions in infants and toddlers with bronchiolitis, and the estimate of effect was highest with tachypnoea (adjusted odds ratio (aOR): 4.68; 95% CI: 2.50–8.76). Chest retractions were associated with hospital admissions (aOR: 2.96; 95% CI: 1.53–5.69) and higher levels of oxygen saturation were protective of admissions in infants and toddlers with bronchiolitis. An SpO₂ of less than 92% was associated with a threefold increase in admissions (aOR: 3.68; 95% CI: 1.67–8.13). Other findings that increased the risk of admission included younger age, male sex, having siblings aged at least 2 years and older (or school-age siblings), presence of a comorbidity, a high fever and a Glasgow Coma Scale <15. The clinical signs listed in Table 3 are reported based on age-based definitions as found in the systematic reviews. Overall, the systematic review associated tachypnoea, chest retractions, SpO₂, Glasgow Coma Scale <15 and temperature >37.8°C with admission, with the list being similar to that in the current standard in the *Pocket book of hospital care for children* (1). The review finding that fever in bronchiolitis

indicates somewhat more severe disease and warrants admission (or at least reassessment and close observation) means a threshold of 38°C in cases of suspected bronchiolitis is significant (180, 183, 184).

Chronic comorbidities substantially increase the likelihood of admission for bronchiolitis in infants and toddlers, with higher risk associated with multiple comorbidities and certain conditions such as Down syndrome, congenital heart disease, cerebral palsy and cystic fibrosis (Table 4). Children with cerebral palsy (aOR: 2.43; 95% CI: 2.54–7.21), cystic fibrosis (aOR: 2.45; 95% CI: 1.36–4.43), Down syndrome (aOR: 2.53; 95% CI: 1.72–3.72) and congenital heart disease (aOR: 3.35; 95% CI: 2.92–3.84) had approximately 2–3 times greater odds of admission as children without these conditions (185). Infants younger than 6 months with any comorbidity were also at increased risk (aOR: 2.67; 95% CI: 2.42–2.94) (182). One study reported the odds of admission increasing when comparing one comorbidity to three comorbidities (aOR: 2.2, for one comorbidity; aOR: 17.6, for three comorbidities) (187).

Table 4. Association of chronic comorbidities and admission with bronchiolitis

Indication for admission – comorbidity	aOR (95% CI)	P value	Sample size	Study
Cerebral palsy	2.43 (2.54)	<0.05	296 200	Murray (185)
Cystic fibrosis	2.45 (1.36)	<0.05		
Down syndrome	2.53 (1.72)	<0.05		
Congenital heart disease	3.35 (2.92)	<0.05		
Comorbidity by <6	2.67 (2.42–2.94)	<0.05	169 726	Hardelid (182)
1 comorbidity	2.2 (1.5–3.2)	<0.001	1 124	Shmueli (187)
2 comorbidities	2.9 (1.5–5.6)	0.002		
3 comorbidities	17.6 (6.7–46.10)	<0.001		

aOR: adjusted odds ratio.

Benefits and desirable effects

Although these admission criteria are derived from studies with a high risk of bias, partly because they were retrospective, the GDG observed that these clinical signs are common-sense clinical assessment findings that can identify children at risk of mortality or who need hospitalization. While most cases of bronchiolitis are self-limiting, these clinical signs provide an objective assessment for identifying those children that may need hospitalization for supportive care.

Harms and undesirable effects

The use of these signs and other measures, such as pulse oximetry, within diagnostic criteria is inconsistent, and they significantly overlap with signs of other conditions, such as pneumonia and sepsis. Since management of bronchiolitis is predominantly supportive, missing the differential diagnosis of pneumonia and sepsis may be harmful.

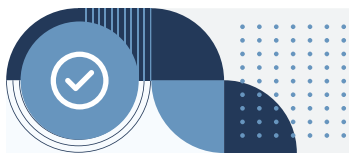
Other considerations

Admission leads to large costs of care, particularly in LMICs. In the United States of America, for children aged less than 2 years, bronchiolitis incurs an annual total inpatient cost of US\$ 1.73 billion.

6.2.3 Research gaps

- There is a need for a well-designed study to determine the factors in the history and physical examination of the infant or child with suspected bronchiolitis requiring hospitalization.

6.3 Use of chest X-ray in the clinical management of bronchiolitis



Recommendation 1

In infants and children aged less than 2 years presenting or hospitalized with suspected bronchiolitis, routine use of chest X-ray is not suggested for the initial diagnosis and management of bronchiolitis.

Conditional recommendation, very low certainty of evidence.

Remarks

- Chest X-ray for hospitalized infants and children with suspected bronchiolitis should only be considered on a case-by-case basis; for example, in the following clinical situations:
 - if there is high suspicion for other alternate diagnosis or underlying cardiac or chronic lung disease related to risk factors elicited on history or physical examination;
 - if a child fails to respond to treatment or persistent symptoms for more than 2 weeks without improvement;
 - if there is suspected bacterial superinfection or recurrent episodes of pneumonia;
 - if there is severe respiratory distress or severe hypoxaemia requiring admission to high care or intensive care;
 - where a complication, such as effusion or pneumothorax, may be suspected; for example, when there are unilateral signs of severe dullness to percussion or absent breath sounds.

6.3.1 Justification

Bronchiolitis is primarily a clinical diagnosis based on history and physical examination findings, including cough, wheezing, tachypnoea and respiratory distress. In most cases, the diagnosis can be made based on these clinical features alone without the need for confirmatory imaging studies. Despite the heterogeneity in the three studies (188-190), the outcomes were consistent that routine chest X-ray (CXR) has no clinical value in typical bronchiolitis, increases the risk of unnecessary antibiotic use and escalates care costs. Driven by these outcomes, along with the evidence that imaging is unlikely to change management outcomes, the GDG decided against routine CXR for suspected bronchiolitis. Routine CXR typically has a low diagnostic yield in children with bronchiolitis, particularly in those without signs of severe respiratory distress or complications.

6.3.2 Summary of evidence

The systematic review identified two prospective studies and one retrospective study conducted in the emergency departments of tertiary hospitals in three high-income countries, with a total of 1194 children aged less than 2 years with clinically diagnosed bronchiolitis (188-190). All the three studies reported on the rates of antibiotic therapy in those that received CXR or change in management after CXR, and, in addition, one prospective cohort study reported on the proportion of CXRs that were inconsistent with clinically diagnosed bronchiolitis. The infants in the included studies had moderate to severe bronchiolitis, with a median age of 2.2 months. The studies had mostly low risk of bias, mainly driven study design (188-190).

The findings from the prospective cohort study suggest that 133 children with typical bronchiolitis would have to undergo radiography to identify one CXR that is suggestive of an alternate diagnosis (190). In addition, it showed that children were five times more likely to receive antibiotic therapy post CXR compared to the intended management plan pre CXR. The other prospective study of 697 children evaluated the relationship between CXR use and antibiotic prescribing. Antibiotic use was significantly higher among those who received a CXR (31%, 108 out of 355) compared with those who did not (8.4%, 28 out of 342). The crude OR was 4.9 (95% CI: 3.1–7.8) (189). The retrospective cohort study assessed 227 children hospitalized for acute bronchiolitis who received a routine CXR and 35 (15.4%) children who received a second CXR in the event of complications. The routine CXRs showed lung abnormalities, specifically distension and/or bronchial wall thickening (62.4%), focal opacity (19.9%) or atelectasis (17.7%). Antibiotic treatment was started in 29 children (12.5%) following use of first CXR (188).

Overall, the three studies suggest that CXR contributed to only partial alteration of treatment of infants hospitalized for acute bronchiolitis, but increased the risk of unnecessary administration of antibiotics. There was low-certainty evidence about the additional value of CXR in management of bronchiolitis in infants aged less than 2 years.

Benefits and desirable effects

The GDG determined that there was trivial benefit from doing routine CXR in all suspected bronchiolitis cases. There was low-certainty evidence that routine CXR was not useful in confirming the diagnosis of bronchiolitis, and that it contributed only partially to alteration of treatment and leads to unnecessary antibiotic use with no clear benefits to patients. The characteristic radiographic findings of bronchiolitis, such as hyperinflation and peri bronchial cuffing, are nonspecific and do not reliably differentiate bronchiolitis from other viral respiratory infections. Nevertheless, the GDG observed that CXR may be beneficial in identifying complications and the promotion of the appropriate use of antibiotics.

Harms and undesirable effects

The GDG judged the undesirable effect as small with very low certainty of evidence. The GDG observed that doing CXR routinely could result in increased use of unnecessary antibiotics, with the likelihood of antimicrobial resistance, and might delay treatment, especially in low-resource settings where X-rays are not easily available at lower-level facilities. Routine CXR may also lead to unnecessary exposure to ionizing radiation which carries potential risks, including an increased lifetime risk of cancer. In addition, routine CXR for children with suspected bronchiolitis can lead to unnecessary health care utilization and diversion of resources, overcrowding in emergency departments, increased costs and potential delays in diagnosis and treatment.

Other considerations

Preferences, values and acceptability: The GDG determined that patients and their caregivers value the main outcome that is being measured and, therefore, on balance, the GDG decided on probably no important uncertainty or variability. However, because of the diverse settings and levels of experience of staff, there may be different degrees of familiarity with clinical practice guidelines. Although patients are likely to accept CXRs, acceptability by clinicians may vary depending on a clinician's confidence and knowledge of the patient's overall clinical condition. Administrators may not want the additional cost of routine X-rays.

Resources required: There was no systematic review of the resources required. However, the GDG observed that the direct costs will be increased, including medical consultation and specialist referrals for unnecessary CXR, which will increase the cost to families. In a cost analysis study of omitting CXR in diagnosis of acute bronchiolitis in 2009, omission of routine CXR saved Canadian dollars 59 per patient, primarily due to savings in X-rays and hospitalizations (191).

Equity: In low-resource settings, there may be long waiting list for CXRs and this might affect ability of patients to receive proper care. The GDG felt that equity is probably reduced in these settings.

Feasibility: The GDG recognized that the feasibility of routine CXR is affected by the limited financial and human resources in LMIC settings and the lack of capacity.

6.3.3 Implementation considerations

- Implementation will require family engagement through effective communication regarding the diagnosis and natural course of the illness. Caregivers should be informed that most cases of bronchiolitis resolve without the need for tests or medications, and that a CXR is unlikely to change the diagnosis and could lead to unnecessary use of antibiotics and exposure to radiation.
- Targeted quality improvement strategies that include the use of clinical champions, implementation of local clinical pathways based on national guidelines, focused educational delivery, educational materials, and audit and feedback may be required to ensure adherence (192).

6.4 Administration of supplemental oxygen through nasal bubble CPAP

No recommendation:

WHO makes no recommendation on the use of nasal bubble CPAP to administer oxygen rather than standard flow oxygen (SFO) through nasal prongs/face mask/headbox for better clinical outcomes.

6.4.1 Justification

The GDG observed that the available evidence was based on a single small study, in which the criteria of the outcomes used had varied definitions that may vary from country to country, and which did not include any of the critical outcomes of interest. The 1 hour of follow-up to assess the need for the escalation of treatment and/or change in the respiratory rate as outcomes did not provide sufficient evidence to make a recommendation. Pulse oximetry would have been a more objective way of measuring clinical improvement in hypoxia, which was not reported in this single study.

The GDG decided not to make a recommendation because of inadequate evidence and knowledge gaps requiring further research.

6.4.2 Summary of evidence

Benefits and desirable effects

The systematic review identified only one RCT with a sample size of 72 infants aged less than 12 months who were randomized to 36 in the bubble CPAP arm and 36 in the standard arm. The primary outcomes measured after 1 hour were change in respiratory rate, Silverman–Anderson score and Modified Paediatric Society of New Zealand Severity Score (193).

The findings from this study were that 14 out of 32 infants in the nasal CPAP group and five out of 35 infants in the standard care group had changes in respiratory rate ≥ 10 . The mean reduction in respiratory rate was greater in the CPAP group (mean: 8.0; standard deviation (SD): 6.0 vs mean: 5.1; SD: 4.0; MD: 2.9; 95% CI: –5.3 to –0.5). Both respiratory severity scores showed greater improvement with CPAP. The Silverman–Anderson score decreased more in the CPAP group (–0.78; SD: 0.87 vs –0.39; SD: 0.73; MD: –0.19; 95% CI: –0.76 to –0.12). Similarly, the Modified Pediatric Society of New Zealand Severity Score showed greater improvement with CPAP (–2.5; SD: 3.0 vs –1.08; SD: 1.300; MD: –1.4; 95% CI: –2.5 to –0.3). There were only two cases that could not tolerate the bubble CPAP (193).

Other considerations

The GDG made no comment on any other aspects of the administration of supplemental oxygen through nasal bubble CPAP on the evidence-to-decision framework.

6.4.3 Research gaps

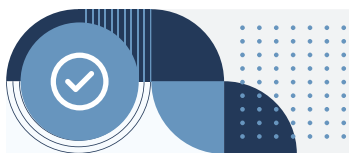
- A large clinical trial with longer follow-up of patients is required comparing bubble CPAP to humidified high-flow nasal cannula (HHFNC) as second-line oxygen escalation therapy for infants and children with acute bronchiolitis where low-flow oxygen therapy is insufficient.

6.5 Humidified high-flow oxygen versus standard low-flow oxygen therapy

6.5.1 Background

Oxygen therapy supplementation is a commonly prescribed intervention when caring for hypoxic or severely ill patients. It often takes the form of standard low-flow oxygen which can deliver up to a maximum of 6 litres per minute or fraction of inspired oxygen (FiO_2) of approximately 0.45 via nasal cannula or prongs. In standard low-flow oxygen therapy, FiO_2 delivery is directly tied to flow rate, and above 6 litres per minute may lead to nasal mucosal irritation and drying of the passages. Humidified high-flow nasal cannula (HHFNC) therapy is an oxygen supply system capable of delivering up to 100% humidified and heated oxygen at a flow rate of up to 60 litres per minute. The high-flow system ensures delivery of a fixed concentration of oxygen and the generation of positive end-expiratory pressure, reduces the work of breathing and clears the nasopharyngeal dead space.

In clinical practice, HHFNC may be used as an alternative to CPAP systems for apnoea of prematurity and post extubating (194). Some of the reported advantages of HHFNC over CPAP include less trauma to the nares, simpler application and making it easier to care for the baby as it gives better access to the baby, including for skin-to-skin contact and feeding (195). Undesired effects include nasal erosion, tiredness when used over long periods, bradycardia, desaturation or high carbon dioxide, and, in addition, “rain out” in the circuit can result in lavage and apnoea (196, 197). Studies have shown HHFNC to be safe as an initial respiratory management, and it is thought to be of benefit for children with bronchiolitis compared with standard oxygen therapy or nasal CPAP (198).



Recommendation 2:

In infants and children aged less than 2 years, hospitalized with acute bronchiolitis with hypoxaemia, HHFNC oxygen is suggested rather than SFO.

Condition recommendation, low certainty of evidence.

Remarks

- In most cases, children with acute bronchiolitis should be started on standard low-flow oxygen, which is very often effective in resolving the hypoxaemia.
- HHFNC oxygen is indicated where, despite standard low-flow oxygen therapy, the infant has persisting hypoxaemia ($\text{SpO}_2 < 90\%$) or severe respiratory distress.

6.5.2 Justification

The recommendation in favour of HHFNC was driven by the moderate certainty of evidence of benefit for the resolution of symptoms/signs such as respiratory rate, heart rate and oxygen saturation, which the GDG considered to be critical outcomes. The GDG also considered the reduction of 0.85 of a day of length of stay in ICU as a clinically important outcome in a situation of a peak RSV epidemic, when there are often too many infants that need hospitalization, as this may save the lives of those who are waiting for hospital beds. Consequently, the GDG issued a conditional recommendation for high-flow oxygen rather than SFO. However, the GDG also noted that if the patients started on SFO have responded well, they do not need to be changed to HHFNC.

6.5.3 Summary of evidence

The systematic review identified seven RCTs comparing HHFNC and SFO conducted in six countries in Africa, Asia and Europe, with a total of 2197 infants and children with moderate to severe bronchiolitis (198-204). The HHFNC studies titrated the fraction of inspired oxygen used to maintain oxygen saturation in a pre-specified range, and the SFO used a variety of delivery methods including nasal prongs, face masks and headbox oxygen. All seven studies reported on length of hospital stay and resolution of symptoms, five studies on escalation to the ICU (199, 201-204), and only three studies reported on adverse events (201-203). Although all were RCTs, all studies had some bias driven by lack of masking/blinding to the intervention, and some had crossovers from the SFO to the HHFNC arms due to treatment failure.

The mean length of stay in hospital was lower in infants that received HHFNC compared to SFO (weighted mean difference (WMD): -0.40 days; 95% CI: -1.071 to 0.27; low-certainty evidence), but this finding was not statistically significant. Two studies reported on length of stay in the ICU; the mean length of stay in the ICU was lower among infants who received HHFNC compared to the standard of care, which was SFO (WMD: -0.85 days; 95% CI: -2.39 to 0.69); however, this finding was not statistically significant (200, 202). In five studies, similar findings that there were fewer infants escalated to the ICU in the HHFNC arm than in the SFO arm (risk ratio (RR): 0.94; 95% CI: -0.57 to -1.53, moderate-certainty evidence), although again this was not statistically significant (199, 201, 203, 204). HHFNC was associated with higher rates of symptom resolution compared to standard oxygen therapy (RR: 1.15; 95% CI: 1.08-1.21; four RCTs including 1887 participants). This equates to 117 more children per 1000 achieving treatment success (moderate-certainty evidence) (199-202).

Seven types of adverse events were reported that were experienced by a total of 19 infants, including apnoea (two infants on HHFNC, three on SFO), epistaxis (two infants on SFO), inhaled condensation (one infant on HHFNC), oxygen desaturation (one infant on HHFNC), oxygen tube disconnection (two infants on SFO), pneumothorax (one infant on HHFNC, one infant on SFO) and transient bradycardia (three infants on HHFNC, two infants on SFO) (201-203). When "any adverse event" in the two trial arms were compared, infants that received the HHFNC intervention experienced 57.9% (11 out of 19 infants) adverse events compared to 42.1% (8 out of 19 infants) among those who were administered SFO (201-203).

Overall, there was moderate certainty of evidence of higher treatment success among those receiving HHFNC therapy, and moderate certainty of evidence of the length of stay in ICU and escalation to ICU for further respiratory support, but low certainty of evidence the same length of stay in hospital. There is no evidence to suggest HHFNC brings about substantially more adverse events than SFO.

Benefits and desirable effects

The GDG determined that anticipated desirable effects were moderate, driven by the moderate certainty of evidence of higher treatment success among those receiving HFNC therapy compared to standard low-flow oxygen therapy. There was little impact on the duration of hospital stay, or on the length of stay in ICU and escalation of treatment.

Harms and undesirable effects

The GDG determined that the undesirable effects were small given that there was no serious adverse events recorded when compared to standard low-flow oxygen therapy. Adverse events reported included apnoea, epistaxis, inhaled condensation, oxygen desaturation, oxygen tube disconnection, pneumothorax and transient bradycardia. There were no differences between HHFNC and SFO therapy in terms of the incidence of adverse events: 57.9% in the HHFNC group compared to 42.1% in the SFO group.

Other considerations

Preferences, values and acceptability: The GDG observed that patients and their caregivers will value the main outcomes that are being measured, and, therefore, decided on probably no important uncertainty or variability. Stakeholders, including caregivers and clinicians, are more likely to look at the outcome as acceptable.

Resources required: There was no systematic analysis on the resources required. However, the GDG considered the possible direct costs as more oxygen therapy is required with HHFNC, along with medical devices and overhead costs. However, in one study, HHFNC was found to be cost-effective compared to oxygen by nasal cannula in an infant with bronchiolitis (206).

Equity: The GDG determined that the recommendation will probably increase equity if it is rolled out as a programme.

Feasibility: Additional considerations that the GDG recognized as influencing feasibility include limited financial and human resources in some settings that may hinder the implementation of the intervention. The evidence from contexts and settings where the intervention has been widely deployed suggests that a comprehensive implementation process and a robust therapy protocol need to be integral to initiating and managing HHFNC in all hospital locations, including appropriate patient selection and staff education (207). This intervention requires increased resource requirements, financial requirements, and retraining health care workers. These factors may limit the feasibility of implementation in some LMIC settings.

6.5.4 Implementation considerations

- Ensuring high-quality care and patient safety.
 - Regular assessment and monitoring of patient vitals (breathing frequency, heart rate, SpO₂), device settings (FiO₂, flow), respiratory pattern, breath sounds and comfort level with the device are prerequisites for high-quality care and improved patient safety (207).
 - A written protocol specifying the clinical indications and contraindications to guide the initiation and management of patients while on HHFNC therapy should be in place.
 - Staff should be trained on the HHFNC protocol, the device, appropriate flow rates according to weight and age, and on regular monitoring of patients.

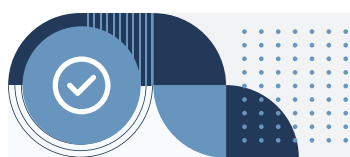
- Health system considerations.
 - The cost of equipment and personnel required, including training, should be considered.
 - Health systems in LMICs are overburdened so there is a need to develop health staff with the appropriate skill mix to support implementation of HHFNC oxygen.

6.6 Nebulized hypertonic (3%) saline versus no hypertonic saline

6.6.1 Background

Bronchiolitis is caused by a viral infection and often leads to hospitalization due to respiratory distress. Bronchiolitis occurs because of the inflammation of the lining of the epithelial cells of the small airways in the lungs causing mucus production, inflammation and cellular necrosis of these cells. It is the inflammation of these cells that can obstruct the airway and, ultimately, result in wheezing. Treatment primarily focuses on supportive care, as there are limited effective pharmacological therapies.

Hypertonic saline (a strong or highly concentrated, usually $\geq 3\%$, sterile saltwater solution) breathed in as a fine mist using a nebulizer may help relieve wheezing and breathing difficulty (208). It is thought to work by reducing airway oedema through osmotic effects, enhancing mucociliary clearance and rehydrating airway surfaces (209). However, studies do not consistently provide evidence for a clear long-term benefit of using nebulized hypertonic saline.



Recommendation 3:

In infants and children aged less than 2 years presenting or hospitalized with bronchiolitis, a trial of nebulized hypertonic saline (3%) is suggested rather than no nebulization.

Conditional recommendation, low certainty of evidence.

Remarks

- The majority of studies used a comparison with normal saline which may underestimate the effect of hypertonic saline.
- A trial of nebulized hypertonic saline in an individual patient may aid in clearance of secretions, but might not always be effective.
- Health facilities experience overcrowding during peak RSV season, making any reduction in the length of hospital stays through the use of nebulized hypertonic saline especially valuable.
- Health care workers should be aware that hypertonic saline can induce bronchoconstriction and may often require co-administration with bronchodilators.

6.6.2 Justification

The recommendation for nebulized hypertonic saline was mainly driven by a modest reduction in the length of stay among infants hospitalized with acute bronchiolitis, reduction in the risk of hospitalization among outpatients and emergency department patients, and the earlier resolution of symptoms. A reduction of hospital stay for even less than 1 day was considered clinically significant, particularly in RSV epidemic situations, although the certainty of the evidence was low to very low for all outcomes, mainly due to inconsistency and risk of bias. In addition, nebulized hypertonic saline seems to be cost-effective and a safe treatment in infants with bronchiolitis, with only minor and spontaneously resolved adverse events.

6.6.3 Summary of evidence

A systematic review identified 38 RCTs conducted over 20 years in Africa, the Americas, Asia and Europe, involving a total of 5114 infants and children with bronchiolitis (208). Study sample sizes ranged from 40 to 772, and patients were followed up for 60 minutes to 72 hours for primary outcomes. All the trials used nebulized hypertonic saline alone, or in conjunction with bronchodilators as an active intervention, and nebulized normal (0.9%) saline or standard treatment as a comparator in children under 24 months with acute bronchiolitis. The primary outcome was length of hospital stay for inpatient trials or rate of hospitalization for the outpatient or emergency department trials. Other study outcomes included were duration of oxygen therapy, resolution of symptoms and adverse events (208).

Hospitalized infants treated with nebulized hypertonic saline may have a shorter mean length of hospital stay compared to those treated with nebulized normal (0.9%) saline or standard care (MD: -0.46 days; 95% CI: -0.56 to -0.36, low-certainty evidence). The risk of hospitalization was reduced by 90 per 1000 patients for nebulized hypertonic saline compared with nebulized normal saline plus salbutamol and was reduced by 112 when compared to normal saline alone. Infants and children who received hypertonic saline had a lower number of days to resolution of wheezing (lower by 1.16 days), cough (lower by 0.92 days), pulmonary crackles (lower by 1.42 days) compared to those who received normal saline (low-certainty evidence). The 13 trials included in the review (2792 infants; 1479 treated with hypertonic saline, of which 416 (28%) were co-administered with bronchodilators and 1063 (72%) treated with hypertonic saline alone) reported at least one adverse event, such as worsening cough, agitation, bronchospasm, bradycardia, desaturation, vomiting and diarrhoea, most of which were mild and resolved spontaneously (low-certainty evidence) (208).

Overall, nebulized hypertonic saline may modestly reduce the length of stay among infants hospitalized with acute bronchiolitis and may slightly reduce time to the resolution of symptoms. Treatment with nebulized hypertonic saline may also reduce the risk of hospitalization among outpatients and emergency department patients. Nebulized hypertonic saline seems to be a safe treatment in infants with bronchiolitis, with only minor and spontaneously resolved adverse events. The certainty of the evidence was low to very low for all outcomes, mainly due to inconsistency and risk of bias.

Benefits and desirable effects

The GDG judged the desirable effects of nebulized hypertonic (>3%) saline over 0.9% saline as moderate, driven by a clinically significant reduction in length of hospital stay, and reduced hospitalization and readmission rates. The GDG noted that that in peak RSV epidemic season, a reduction of hospital stay for even less than 1 day should be considered significant, depending on the setting. In addition, nebulized hypertonic saline contributed to earlier resolution of wheezing, cough and pulmonary crackles when compared to the other interventions. The duration of oxygen therapy was also reduced by 15 minutes in the hypertonic saline group, although this was not statistically significant.

Harms and undesirable effects

The GDG judged the undesirable effects as trivial, and although there were more adverse effects in the hypertonic saline treatment arm (4.6%; 67 out of 1449 patients) than standard care (2.3%; 30 out of 1291 patients), this difference was not statistically significant. Nebulized hypertonic saline seems to be a safe treatment in infants with bronchiolitis, with only minor and spontaneously resolved adverse events. In addition, the GDG noted that the use of hypertonic saline might lead to 13 more adverse events per 1000 children, although the confidence interval ranged from eight fewer children to 64 more children.

Other considerations

Preferences, values and acceptability: There might be variability in how caregivers will value this main outcome. Nebulization involves placing a mask on the child's face and this might raise anxiety levels in some parents; however, most stakeholders will probably accept the intervention.

Resource required: By reducing hospital stay and admission rate, the use of hypertonic saline may be cost saving and, therefore, the GDG decided that the resources required were negligible. In one study, the nebulized 3% hypertonic solution was cost-effective in the inpatient treatment of an infant with bronchiolitis (210).

Equity: Equity is most likely to vary depending on the context, such as the availability of nebulizers/masks, infrastructure, etc., and patients in rural settings might be disadvantaged. Hypertonic saline may not be available, even though it is not costly. The supply chain issues for nebulizers/masks and the quality of nebulizers during epidemics, as well as the education of parents, will affect implementation.

Feasibility: The considerations looked at when assessing the feasibility of implementation included stock-outs of nebulizers and the quality of nebulizers, but the GDG was of the opinion that implementation was probably feasible.

6.6.4 Implementation considerations

- Caregivers need to be educated and counselled on the value of nebulization.
- When recommending the use of hypertonic saline nebulization on a large scale, especially during bronchiolitis epidemics, availability and stocks need to be addressed.
- The rules for discharge from hospital vary by hospital and by country, and even by age of the admitted infant, meaning that hospital stays for a 3-month-old may be different from those of a 15-month-old.

6.6.5 Research gaps

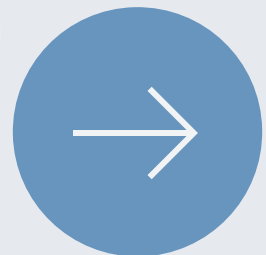
- Future studies should examine older populations, including children aged between 12 months and 24 months, implement post-randomization limitation of bias (masking), and report the adverse events experienced by participant level.
- Other research questions include:
 - what is the number of doses required for resolution of symptoms and signs?
 - what is the optimal dosing regimen for nebulized therapy in bronchiolitis?
 - what is the age distribution of bronchiolitis cases?
 - what is the length of stay for children aged less than 6 months who are nebulized?
 - what is the effect of nebulized therapy on feeding tolerance and oral intake in infants with bronchiolitis?
 - what clinical signs or diagnostic approaches are best in differentiating bronchiolitis from asthma?

7. Monitoring and evaluation



Monitoring and evaluation will be built into the dissemination and implementation process to provide data and information on uptake, implementation and impact of this guideline. In collaboration with the monitoring and evaluation team of the Department of Sexual, Reproductive, Maternal, Child and Adolescent Health and Ageing, WHO, the data on country- and regional-level adoption of the recommendations will be monitored and evaluated in the short-to-medium term in Member States, through the WHO sexual, reproductive, maternal, newborn, child and adolescent health policy survey (211). This survey is conducted every 4 years.

8. Dissemination, adaptation, implementation and update of the guideline



As this is a global guideline, Member States are expected to adapt the recommendations to their setting, considering feasibility, resource availability and other considerations at the national and subnational level. WHO regional and country offices can assist with the adaptation processes. The dissemination and implementation of this guideline is to be considered by all stakeholders involved in the provision of care for children and adolescents.

8.1 Derivative products

These recommendations are part of the consolidated evidence for the updates of the *Pocket book of hospital care for children (1)* and the *Integrated management of childhood illness (IMCI) chart booklet (3)*, which are the main derivatives of these recommendations. In addition to these publications, digital adaptation kits, training manuals, assessment tools and job aids will be updated to provide consistent implementation guidance.

8.2 Dissemination

An executive summary containing the recommendations, remarks, implementation considerations and research priorities will be prepared for public dissemination. WHO will also develop derivative tools to aid understanding and adaptation of these recommendations to local contexts.

The recommendations and derivative tools will be disseminated through the WHO website, and targeted webinars, through a broad network of international partners, including WHO regional and country offices, ministries of health, professional organizations, universities, WHO collaborating centres, other United Nations agencies and nongovernmental organizations, among others. Updated recommendations are also routinely disseminated during meetings and scientific conferences attended by WHO representatives.

The executive summary and recommendations from this publication will be translated into the official United Nations languages for dissemination through the WHO regional and country offices and during meetings organized by, or attended by, staff of the Department of Sexual, Reproductive, Maternal, Child and Adolescent Health and Ageing and the Department of Noncommunicable Diseases and Mental Health, WHO. Technical assistance will be provided to any WHO regional office wanting to translate these recommendations into any of their languages. In addition, the publication of journal articles presenting the recommendations and key implementation considerations will be considered in compliance with WHO's open access and copyright policies.

To ensure that these recommendations have a positive impact on asthma in children and adolescents and on bronchiolitis in infants and young children, coordinated action between WHO, other agencies, national departments of health and other key stakeholders is needed. National and subnational working groups should assess current national guidelines and protocols and determine whether development of new guidelines or updating of existing guidelines is required in line with these new WHO recommendations. WHO staff at the headquarters, regional and country levels, as well as international agency partners and international professional societies and national professional associations, can support national stakeholders in developing or revising existing national guidelines or protocols, and optimizing their implementation.

In the context of humanitarian emergencies, the adaptation of the current recommendations should consider their integration and alignment with other response strategies. Context-specific tools and toolkits may be required in addition to standard tools to support the implementation of the recommendation in humanitarian emergencies.

8.3 Adaptation

Public child health programmes require supportive policies and health care services that enable the proper access to quality services and medicines. WHO headquarters will work with WHO regional and country offices to assist with the introduction, orientation and adaptation processes. For most Member States, an intensive and resource-heavy adaptation process will not be necessary. All efforts were made to ensure that, where possible, the recommendations align with accepted clinical decision-making algorithms. However, even if the adaptation process is not intensive, ministries of health will have to update their national standard treatment guidelines based on the *Pocket book of hospital care for children (1)* and the *Integrated management of childhood illness (IMCI) chart booklet (3)*, update their training curricula and also develop plans for building capacity for health care workers on the updated clinical algorithms.

8.4 Implementation

There are a number of important implementation considerations that have been highlighted in the guideline for end-users as they implement these recommendations. These are necessary in order to provide accessible, high-quality health services for the management of asthma in children and adolescents, and the management of bronchiolitis in infants and young children. These considerations may assist policy-makers, clinicians and other stakeholders to better prepare for implementation. Member States and other end-users need to ensure that the necessary policies, regulations, infrastructure, supplies and skilled and competent personnel are in place.

Local professional societies and training institutions have an important role in implementation, and an all-inclusive and participatory process should be encouraged. National health systems must ensure that supplies of medicines and commodities such as suitable monitoring are available in health care facilities where services are provided. In settings where a new practice is introduced (or where recommended practices are changed), additional training and monitoring may be required. In contexts with high rates of personnel turnover, pre-service training and regular opportunities for ongoing training and competency assessment are particularly important.

8.4 Updating

The WHO Secretariat for these recommendations will continue to follow developments in the research on the prevention, diagnosis and management of asthma and bronchiolitis, particularly for questions for which the certainty of evidence was found to be low or very low. If new evidence emerges, or other important considerations arise that may impact the current recommendations, the Department of Sexual, Reproductive, Maternal, Child and Adolescent Health and Ageing, WHO, will coordinate an update of this guideline, following the procedures outlined in the *WHO handbook for guideline development (15)*.

Unless new evidence necessitates an earlier review, at 5 years from publication of this guideline, the Department of Sexual, Reproductive, Maternal, Child and Adolescent Health and Ageing, together with its internal partners, will conduct systematic reviews of the relevant evidence and appraise the need for updating or revalidating the current guideline. WHO will seek stakeholder input on the scope of the updated guideline, as new interventions and considerations emerge.

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Annex 1. Management of conflicts of interest for the Guideline Development Group

In accordance with WHO policies and procedures for guideline development, all members of the Guideline Development Group (GDG) were required to declare any potential conflicts of interest prior to their involvement. This was done through the completion and electronic submission of the standard WHO Declaration of Interests (DOI) form.

In addition, all GDG members, methodologists and external reviewers signed confidentiality agreements to ensure the integrity and impartiality of the guideline development process.

All submitted DOI forms were reviewed by the WHO Secretariat. Any declared interests were assessed and managed in line with the procedures outlined in the *WHO handbook for guideline development*⁶ and WHO's Declarations of interests for WHO experts.⁷ Where a declared interest was considered potentially significant, appropriate measures were taken. These included conditional participation, partial exclusion from specific discussions or full exclusion from the guideline process, depending on the nature and extent of the interest.

At the beginning of each GDG meeting, participants were asked to declare any new or updated interests. In cases where a GDG member had contributed to a study that informed the evidence for a specific recommendation, that member was recused from the relevant discussions and decision-making, as documented in the meeting records.

Table A1.1. Summary of the declared interests of the GDG

Name	Affiliation	WHO Region	Expertise	Declared interest(s)
Mohamed Ahmed Abdullah	University of Khartoum, Khartoum, Sudan	Regional Office for the Eastern Mediterranean	Paediatric endocrinologist	None declared
Shams El Arifeen ⁸	International Centre for Diarrhoeal Disease Research, Dhaka, Bangladesh	Regional Office for South-East Asia	Paediatric infectious diseases	Yes ⁹
Hesham Ben Masaud	General Practice Council, Libyan Board of Medical Specialties, Tripoli, Libya	Regional Office for the Eastern Mediterranean	Paediatric programme specialist	None declared

6 WHO handbook for guideline development. Geneva: World Health Organization; 2014 (<https://iris.who.int/handle/10665/145714>).

7 Declaration of interests for WHO experts, <https://www.who.int/publications/m/item/declaration-of-interests-for-who-experts>.

8 Shams El Arifeen and Michael (Mike) English chaired the GDG meetings.

9 The declared intellectual conflict of interest was not considered significant enough to pose any risk to the guideline development process or to reduce its credibility.

Gillian Birchwood	The Queen Elizabeth Hospital, Bridgetown, Barbados	Regional Office for the Americas	Neonatologist	None declared
Trevor Duke	University of Melbourne, Faculty of Medicine, Dentistry and Health Sciences, Melbourne, Australia	Regional Office for the Western Pacific	Intensive care, oxygen systems and neonatologist	None declared
Michael (Mike) English ³	University of Oxford, Oxford, United Kingdom of Great Britain and Northern Ireland	Regional Office for Europe	Paediatrics and child health; quality of care	Yes ⁴
Grace Irimu	University of Nairobi, Nairobi Kenya	Regional Office for Africa	Neonatology; paediatric quality of care	None declared ¹⁰
Phillip Kasirye	Makerere University College of Health Sciences, Kampala, Uganda	Regional Office for Africa	Paediatric haematologist; sickle-cell disease	None declared
Shanzida Khatun	National Institute of Advanced Nursing Education and Research, Dhaka, Bangladesh	Regional Office for South-East Asia	Child health nursing	None declared
Ajay Khera	Engender Health, New Delhi, India	Regional Office for South-East Asia	Community and gender perspective	None declared
Olga Komarova	National Medical Research Center of Children's Health, Moscow, Russian Federation	Regional Office for Europe	Paediatric quality of care, adolescent health	None declared
Ida Safitri Laksono	Universitas Gadjah Mada, Yogyakarta, Indonesia	Regional Office for South-East Asia	Paediatric infectious disease, dengue	None declared
Rakesh Lodha	All India Institute of Medical Sciences, New Delhi, India	Regional Office for South-East Asia	Paediatric intensive care, pulmonology and infectious diseases	None declared
Refiloe Masekela	University of KwaZulu Natal, Durban, South Africa	Regional Office for Africa	Paediatric pulmonologist	None declared

10 Conflict of interest management was not applicable in this case.

Hilda Mujuru	University of Zimbabwe, Harare, Zimbabwe	Regional Office for Africa	Paediatric infectious diseases – HIV, Integrated Management for Newborn and Childhood Illnesses (IMNCI)	None declared
Mary Paiva	Buka General Hospital, National Department of Health), Buka, Autonomous Region of Bougainville, Papua New Guinea	Regional Office for the Western Pacific	General paediatrics, community, social paediatrics, district-level rural health services	None declared
Abdulkarim Rasa'a	Faculty of Medicine Sana'a University, Sana'a, Yemen	Regional Office for the Eastern Mediterranean	Paediatric infectious disease	None declared
Sergey Sargsyan	Arabkir Medical Centre. Yerevan State Medical University, Yerevan, Armenia	Regional Office for Europe	Paediatrics and adolescent preventive medicine	None declared
Dheeraj Shah	University College of Medical Sciences and GTB Hospital, New Delhi, India	Regional Office for South-East Asia	Pediatric gastroenterology and nutrition	None declared
Gezahegn Nekatibeb Techane	Debre Berhan University, Debre Birhan, Ethiopia	Regional Office for Africa	General paediatrician; paediatric quality of care specialist	None declared
Dai Yaohua	Capital Institute of Paediatrics, Beijing, China	Regional Office for the Western Pacific	Child health nutrition, early childhood development	Gezahegn Nekatibeb Techane

Annex 2. Prioritized PICO questions on the clinical management of asthma in children and adolescents

1. In children and adolescents <19 years of age with acute asthma exacerbation, should inhaled/nebulized ipratropium be added to standard first-line treatment with inhaled/nebulized short-acting beta2 agonists (SABA) and systemic corticosteroids?

Population

Children and adolescents <19 years of age with acute asthma exacerbation

Intervention

- Inhaled/nebulized SABA and systemic steroids + nebulized ipratropium

Comparator

- Inhaled/nebulized SABA and systemic steroids

Subgroups

- By severity: mild-moderate or severe
- Immediately started with first-line treatment
- Started if poor response to first-line treatment

Outcomes

- Morbidity (symptoms, hospitalization, school absence)
- Escalation of care (ventilation/PICU admission)
- Length of stay (in emergency department or hospital)
- Mortality due to exacerbation
- Adverse events
- Lung function (change in FEV₁, PEF) – less important

FEV₁: forced expiratory volume in 1 second; PEF: peak expiratory flow; PICU: paediatric intensive care unit; SABA: short-acting beta2 agonists.

2. In children and adolescents <19 years of age with acute asthma exacerbation, should IV magnesium sulfate be added to standard first-line treatment with inhaled/nebulized SABA and systemic corticosteroids?

Population

Children and adolescents <19 years of age with acute asthma exacerbation

Intervention

- Inhaled/nebulized SABA and systemic steroids + IV magnesium sulfate

Comparator

- Inhaled/nebulized SABA and systemic steroids

Subgroups

- Immediately started with first-line treatment
- Started if poor response to first-line treatment

Outcomes

- | | |
|--|--|
| <ul style="list-style-type: none"> • Morbidity (symptoms, hospitalization, school absence) • Escalation of care (ventilation/PICU admission) • Length of stay (in emergency department or hospital) | <ul style="list-style-type: none"> • Adverse events • Mortality due to exacerbation • Lung function (change in FEV₁, PEF) – less important |
|--|--|

FEV₁: forced expiratory volume in 1 second; IV: intravenous; PEF: peak expiratory flow; PICU: paediatric intensive care unit; SABA: short-acting beta₂ agonist.

3. In children and adolescents <19 years of age with acute severe asthma exacerbation, should IV aminophylline be added to standard first-line treatment with inhaled/nebulized SABA and systemic corticosteroids?

Population

Children and adolescents <19 years of age with acute severe asthma exacerbation

Intervention

- Inhaled/nebulized SABA and systemic steroids + IV aminophylline

Comparator

- Inhaled/nebulized SABA and systemic steroids

Subgroups

- Immediately started with first-line treatment
- Started if poor response to first-line treatment

Outcomes

- | | |
|--|--|
| <ul style="list-style-type: none"> Morbidity (symptoms, hospitalization, school absence) Escalation of care (ventilation/PICU admission) Length of stay (in emergency department or hospital) | <ul style="list-style-type: none"> Adverse events Mortality due to exacerbation Lung function (change in FEV₁, PEF) – less important |
|--|--|

FEV₁: forced expiratory volume in 1 second; IV: intravenous; PEF: peak expiratory flow; PICU: paediatric intensive care unit; SABA: short-acting beta₂ agonist.

4. In children and adolescents <19 years of age with acute severe asthma exacerbation, should subcutaneous adrenaline be added to standard first-line treatment with inhaled/nebulized SABA and systemic corticosteroids?

Population

Children and adolescents <19 years of age with acute severe asthma exacerbation

Intervention

- Inhaled/nebulized SABA and systemic steroids + subcutaneous adrenaline

Comparator

- Inhaled/nebulized SABA and systemic steroids

Outcomes

- | | |
|--|--|
| <ul style="list-style-type: none"> Morbidity (symptoms, hospitalization, school absence) Escalation of care (ventilation/PICU admission) Length of stay (in emergency department or hospital) | <ul style="list-style-type: none"> Adverse events Mortality due to exacerbation Lung function (change in FEV₁, PEF) – less important |
|--|--|

FEV₁: forced expiratory volume in 1 second; IV: intravenous; PEF: peak expiratory flow; PICU: paediatric intensive care unit; SABA: short-acting beta₂ agonist.

5. In children and adolescents <19 years of age with acute severe asthma exacerbation, should IV SABA (e.g. salbutamol) be added to standard first-line therapy with inhaled/nebulized SABA and systemic corticosteroids?

Population

Children and adolescents <19 years of age with acute severe asthma exacerbation

Intervention

- Inhaled/nebulized SABA and systemic steroids + IV SABA

Comparator

- Inhaled/nebulized SABA and systemic steroids

Outcomes

- Morbidity (symptoms, hospitalization, school absence)
- Escalation of care (ventilation/PICU admission)
- Length of stay (in emergency department or hospital)
- Adverse events
- Mortality due to exacerbation
- Lung function (change in FEV₁, PEF) – less important

FEV₁: forced expiratory volume in 1 second; IV: intravenous; PEF: peak expiratory flow; PICU: paediatric intensive care unit; SABA: short-acting beta₂ agonist.

6. In children and adolescents <19 years of age with acute severe asthma exacerbation, which second-line therapy is most effective?

Population

Children and adolescents <19 years of age with acute severe asthma exacerbation

Intervention

- IV aminophylline
- IV SABA
- IV magnesium sulfate
- Subcutaneous adrenaline

Comparator

- Versus each other

Outcomes

- Morbidity (symptoms, hospitalization, school absence)
- Escalation of care (ventilation/PICU admission)
- Length of stay (in emergency department or hospital)
- Mortality due to exacerbation
- Adverse events
- Lung function (change in FEV₁, PEF) – less important

FEV₁: forced expiratory volume in 1 second; IV: intravenous; PEF: peak expiratory flow; PICU: paediatric intensive care unit; SABA: short-acting beta₂ agonist.

7. In children and adolescents <19 years of age with acute severe asthma exacerbation, which second-line therapy is most effective?

Population

Children and adolescents <19 years of age with asthma

Subgroups

- Age: ≤5 years
- Age: 6–11 years
- Age: ≥12 years

Intervention

- Combination inhaler (steroid plus fast-onset LABA)

Comparator

- Separate steroid (maintenance) and SABA

Outcomes

- | | |
|---|---|
| • Morbidity (symptoms, exacerbations, school absence) | • Daily inhaled steroid dose |
| • Emergency care visits or hospital admissions | • Adverse events |
| • Discontinuation | • Lung function (change in FEV ₁ , PEF) – less important |

FEV₁: forced expiratory volume in 1 second; LABA: long-acting beta₂ agonist; PEF: peak expiratory flow; SABA: short-acting beta₂ agonist.

Annex 3. Prioritized PICO questions on the clinical management of bronchiolitis in infants and young children

1. In infants and children aged less than 2 years with suspected bronchiolitis, what factors in the history and physical examination contribute to a differential diagnosis of acute bronchiolitis?

Population

Children aged less than 2 years with suspected bronchiolitis

Intervention

- Combination of signs and symptoms

Comparator

- Current practice

Outcomes

- Correct diagnosis
- Sensitivity and specificity
- Appropriate use of antibiotics

2. In infants and children aged less than 2 years presenting to a primary care setting with suspected bronchiolitis, what are the clinical indicators for admission?

Population

Children aged less than 2 years with suspected bronchiolitis

Intervention

- Combination of severe signs and symptoms (emergency signs – cyanosis, severe respiratory distress, severe chest indrawing, hypoxaemia (SpO₂ <90%))

Comparator

- Current practice

Outcomes

- Priority outcomes: hospital admission
- Prolonged stay in hospital
- Intensive care unit admission
- Mortality

SpO₂: peripheral capillary oxygen saturation.

3. In infants and children aged less than 2 years presenting to hospital, or hospitalized with acute bronchiolitis, does performing a routine chest X-ray beneficially change clinical management or treatment outcomes?

Population

Children aged less than 2 years with bronchiolitis

Intervention

- Clinical signs and symptoms + chest X-ray

Comparator

- Only clinical signs and symptoms

Outcomes

- Priority outcomes
- Appropriate use of antibiotics
- Diagnostic accuracy
- Readmission

4. In infants and children aged less than 2 years hospitalized with acute bronchiolitis and hypoxaemia, does administration of supplemental oxygen through nasal bubble continuous positive airway pressure (CPAP) rather than standard flow oxygen through nasal prongs/face mask/headbox oxygen improve clinical outcomes?

Population

Children aged less than 2 years with acute bronchiolitis and hypoxaemia

Intervention

- Supplemental oxygen through bubble CPAP

Comparator

- Standard flow oxygen through nasal prongs/face mask/headbox oxygen

Subgroups

- Severity of illness

Outcomes

- Need for mechanical ventilation
- Length of hospital stay
- Escalation of care to intensive care unit
- Adverse events

CPAP: continuous positive airway pressure.

5. In infants and children aged less than 2 years hospitalized (admitted) with acute bronchiolitis and hypoxaemia, does administration of supplemental oxygen through humidified high-flow oxygen/air via cannula improve clinical outcomes compared to standard flow oxygen?

Population

Children aged less than 2 years with severe bronchiolitis and hypoxaemia or severe respiratory distress despite standard flow rate oxygen therapy

Intervention

- Humidified high-flow oxygen

Comparator

- Standard flow rate oxygen

Outcomes

- Length of hospital stay
- Escalation of care to intensive care unit
- Adverse events
- Resolution of symptoms

6. In infants and children aged less than 2 years presenting to hospital, or hospitalized with acute bronchiolitis, does administration of nebulized hypertonic saline rather than no nebulization improve clinically relevant end-points?

Population

Children aged less than 2 years with bronchiolitis

Intervention

- Hypertonic ($\geq 3\%$) saline solution

Comparator

- No nebulization

Outcomes

- | | |
|--|---|
| <ul style="list-style-type: none"> • Length of stay • Admission or readmission rate • Adverse event | <p>Other outcomes:</p> <ul style="list-style-type: none"> • Duration of oxygen therapy • Resolution of symptoms |
|--|---|

For questions or enquiries about this guideline, please contact:

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